

수소경제와 에너지저장 시스템

(Hydrogen Economy and Energy Storage Systems)

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Hankuk University of Foreign Studies (HUFS)

2. Introduction to Hydrogen (2)

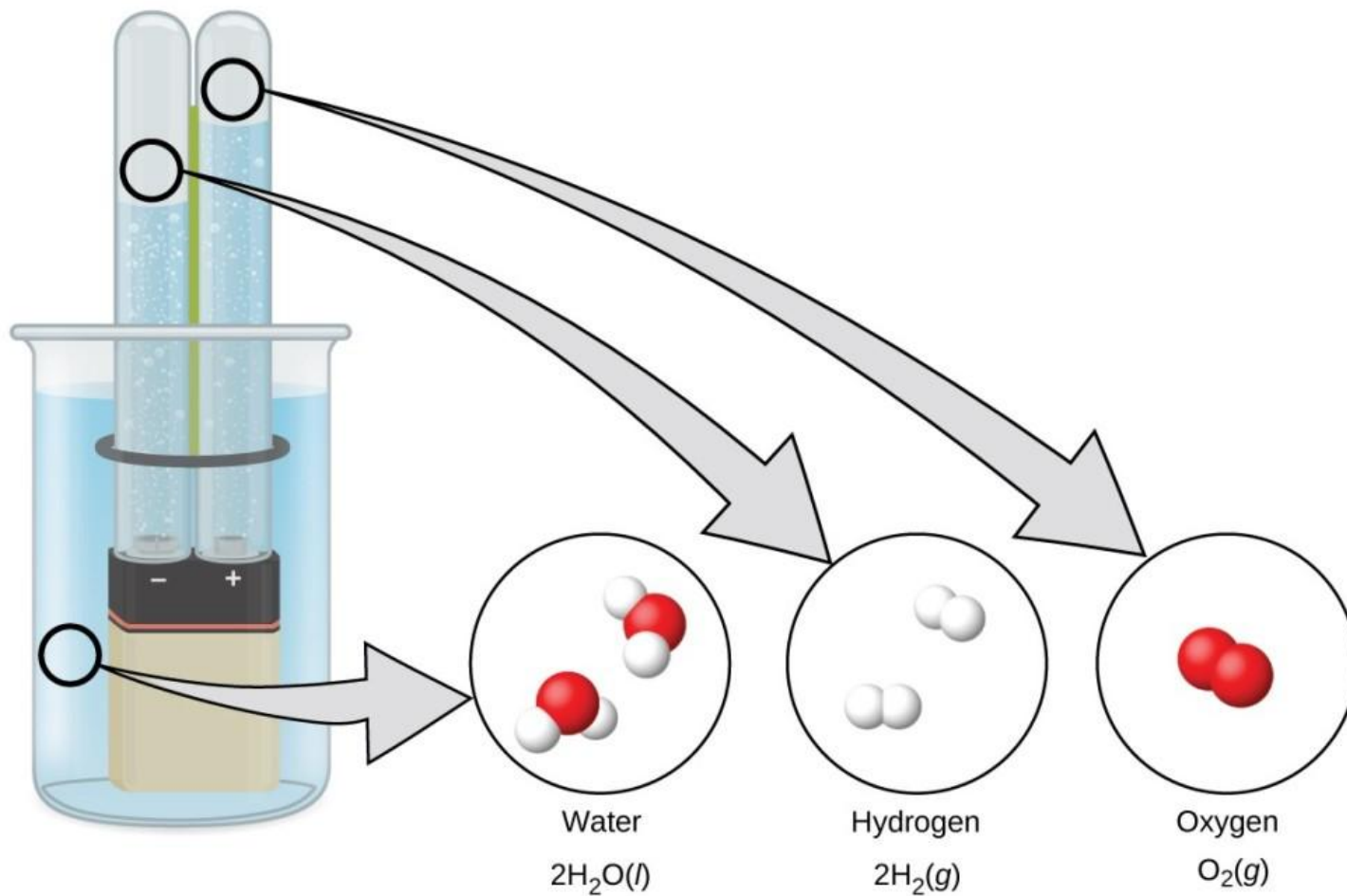
Contents



1. Green Hydrogen Production
2. Hydrogen Storage & Transportation
3. Hydrogen Utilization
4. About Fuel Cell

Water Splitting

□ Electrochemical Reduction of Water

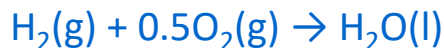


Hydrogen Energy

□ Oxidation of Hydrogen

Exothermic reaction where hydrogen burns with oxygen to produce water/steam and heat

● Higher-Heating Value (HHV)



$$\Delta H^\circ = -285.8 \text{ kJ mol}^{-1} (= \sim 141.8 \text{ MJ kg}^{-1})$$

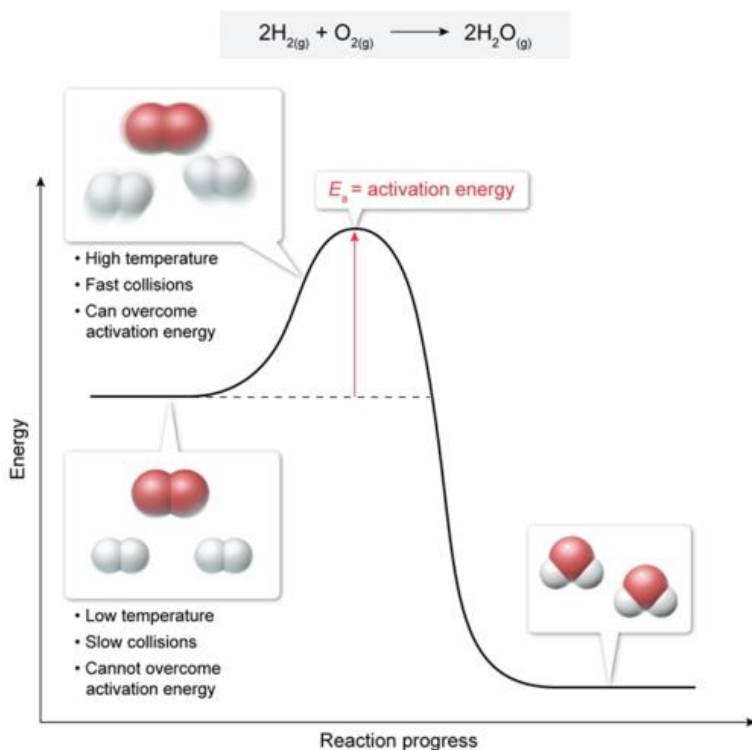
$$\Delta G^\circ = -237.1 \text{ kJ mol}^{-1}$$

● Lower-Heating Value (LHV)



$$\Delta H^\circ = -241.8 \text{ kJ mol}^{-1} (= \sim 120.0 \text{ MJ kg}^{-1})$$

$$\Delta G^\circ = -228.6 \text{ kJ mol}^{-1}$$



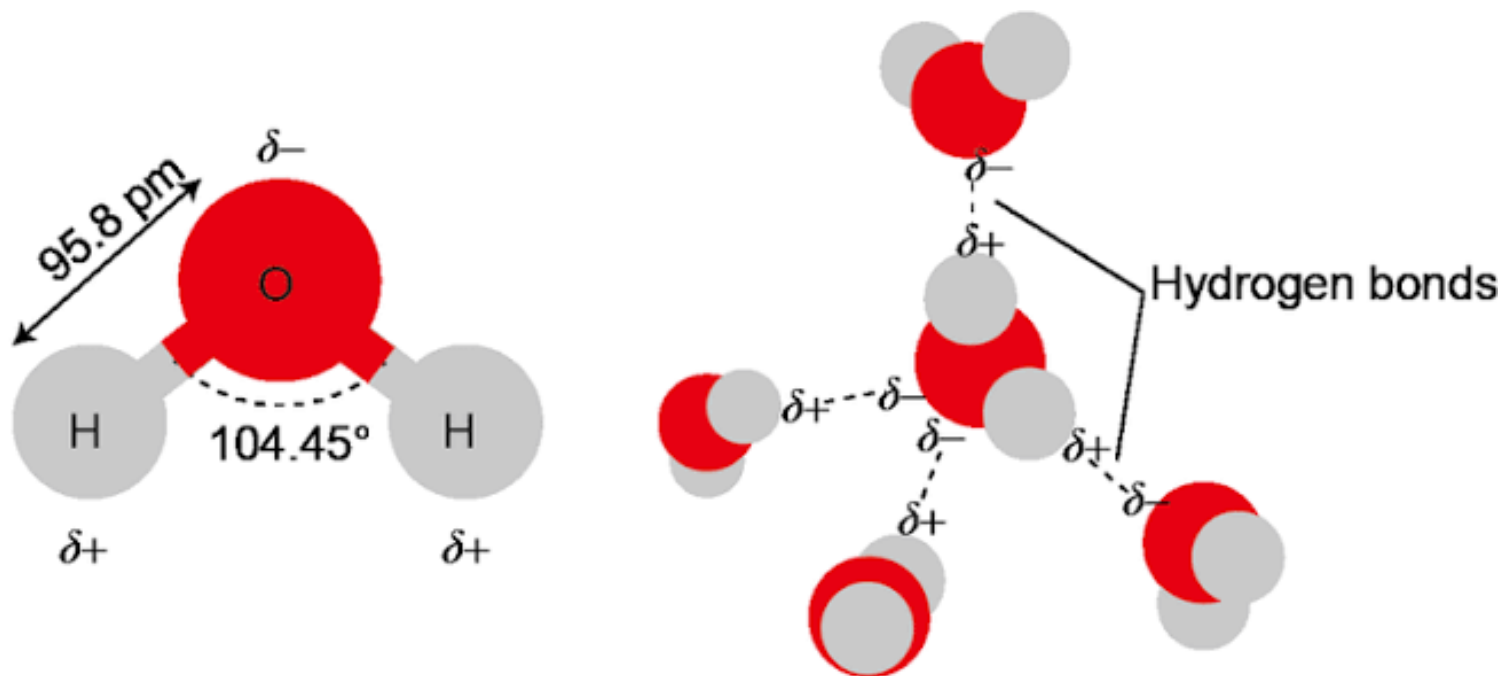
Temp (°C)	ΔH (kJ mol ⁻¹)	ΔG (kJ mol ⁻¹)
25	-285.8	-237.1
50	-285.0	-233.1
100	-283.5	-225.2
150	-243.1	-222.8
200	-243.6	-220.4
300	-244.5	-215.4
400	-245.4	-210.2
500	-246.2	-204.9
600	-247.0	-199.5
700	-247.7	-194.1
800	-248.3	-188.5

Water Splitting

□ Structure of Water

Consists of one oxygen atom joined to two hydrogen atoms by single polar covalent bonds

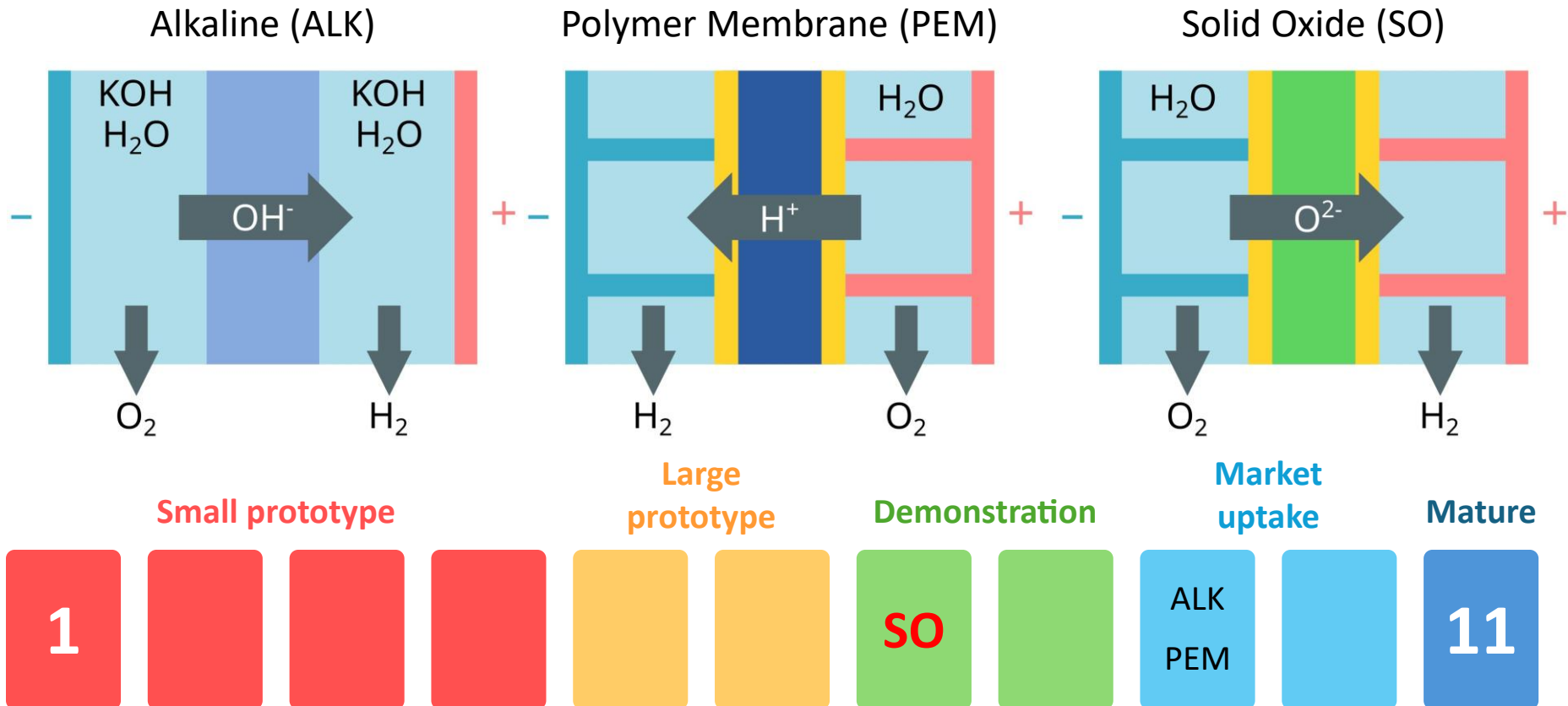
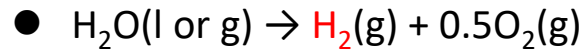
- Bent structure with a bond angle of 104.5° , caused by two lone pairs on the oxygen atom
- O–H bond distance of 95.8 pm ($=0.958 \text{ \AA} = 9.58 \times 10^{-11} \text{ m}$)
- Abnormally high boiling point (100°C), high surface tension (72.8 mN m^{-1} at 20°C), and density anomaly upon freezing



Electrolysis Hydrogen

□ Water (Steam) Electrolysis

A process of using direct electric current to split water into hydrogen and oxygen

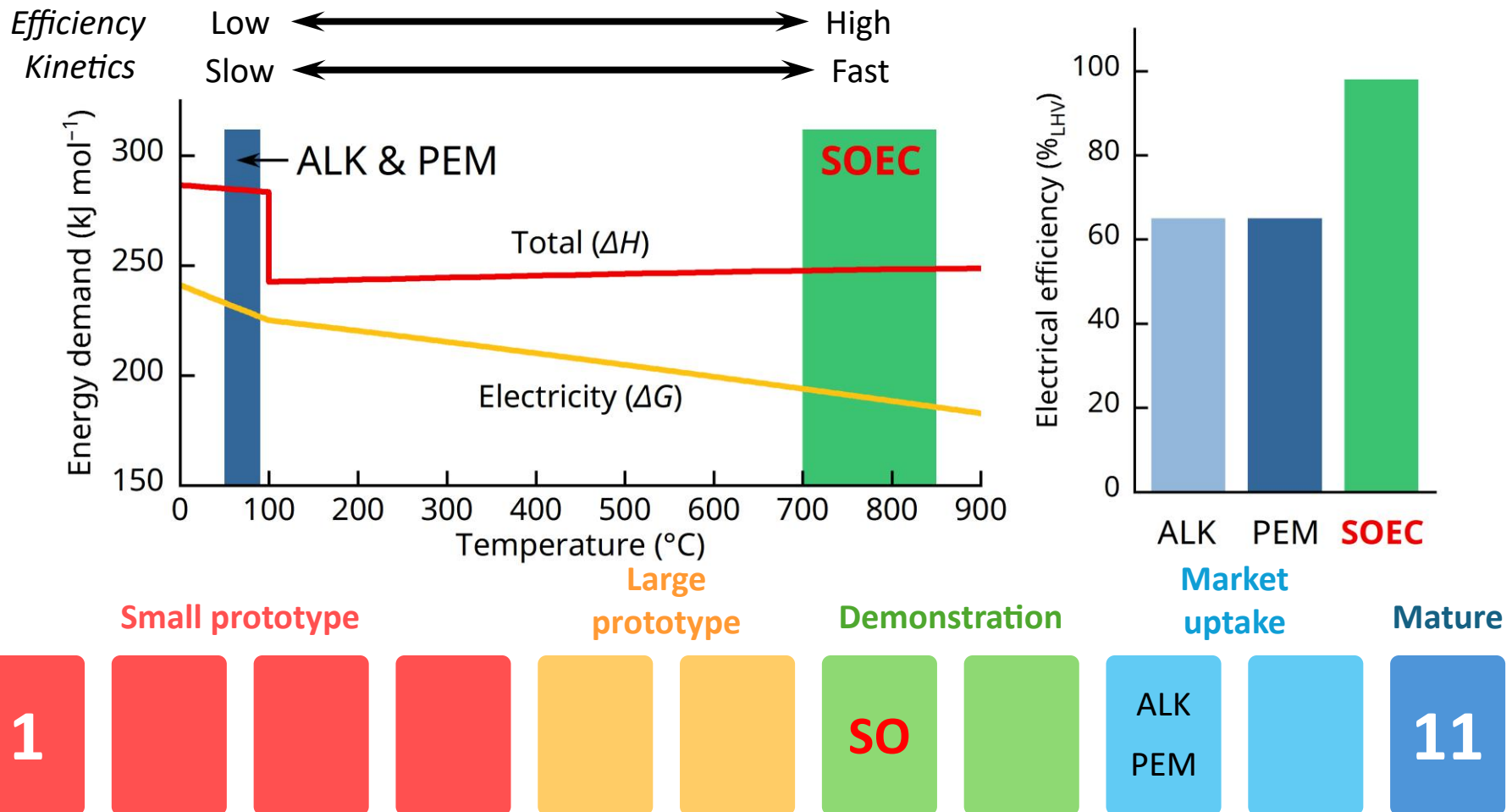


<Source: IEA Electrolyser>

Electrolysis Hydrogen

Water (Steam) Electrolysis

A process of using direct electric current to split water into hydrogen and oxygen



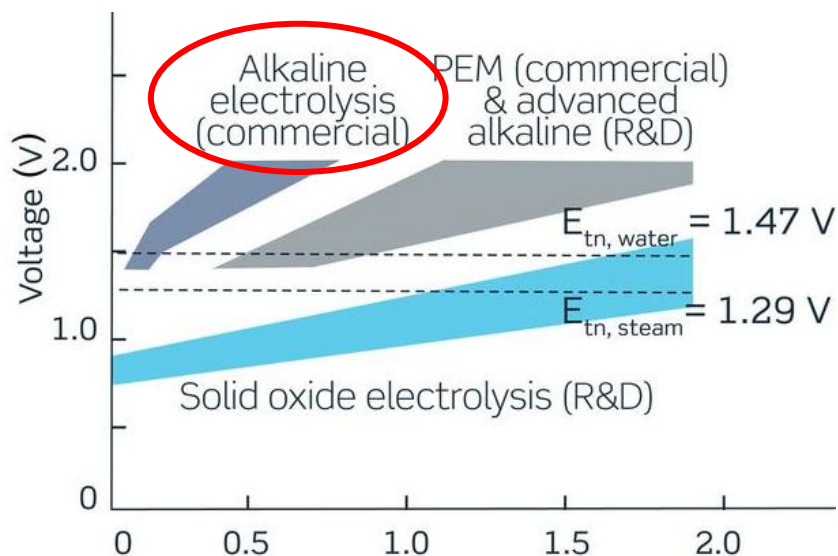
<Source: IEA Electrolyser>

Electrolysis Hydrogen

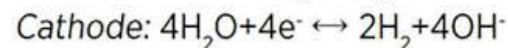
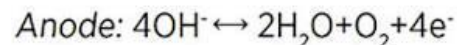
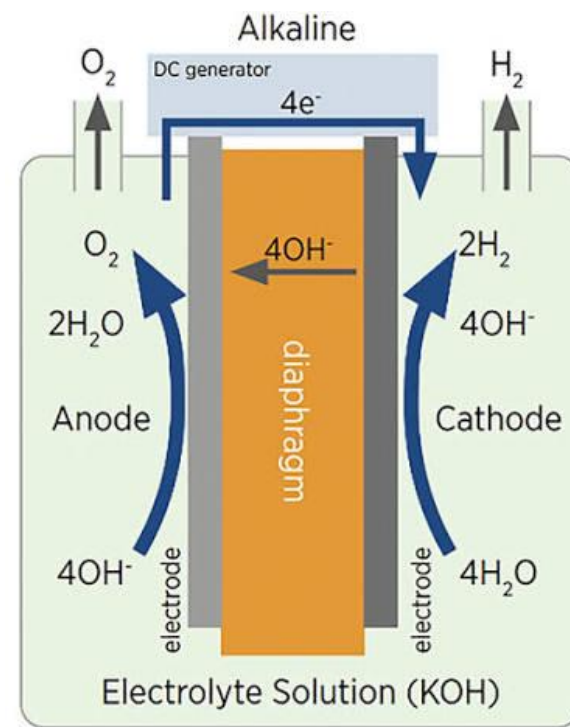
□ Alkaline Water Electrolysis

Characterized by having two electrodes operating in a *liquid alkaline electrolyte*, where a 25–40 wt% KOH or NaOH solution is used

- A highly reliable, mature electrolysis technology with a technology readiness level (TRL) of 9
- High power consumption
- Necessity of corrosion-resistant electrodes (e.g., Ni-based alloy)
- Periodic replenishment of electrolyte



Science, **370**(6513) (2020) aba6118



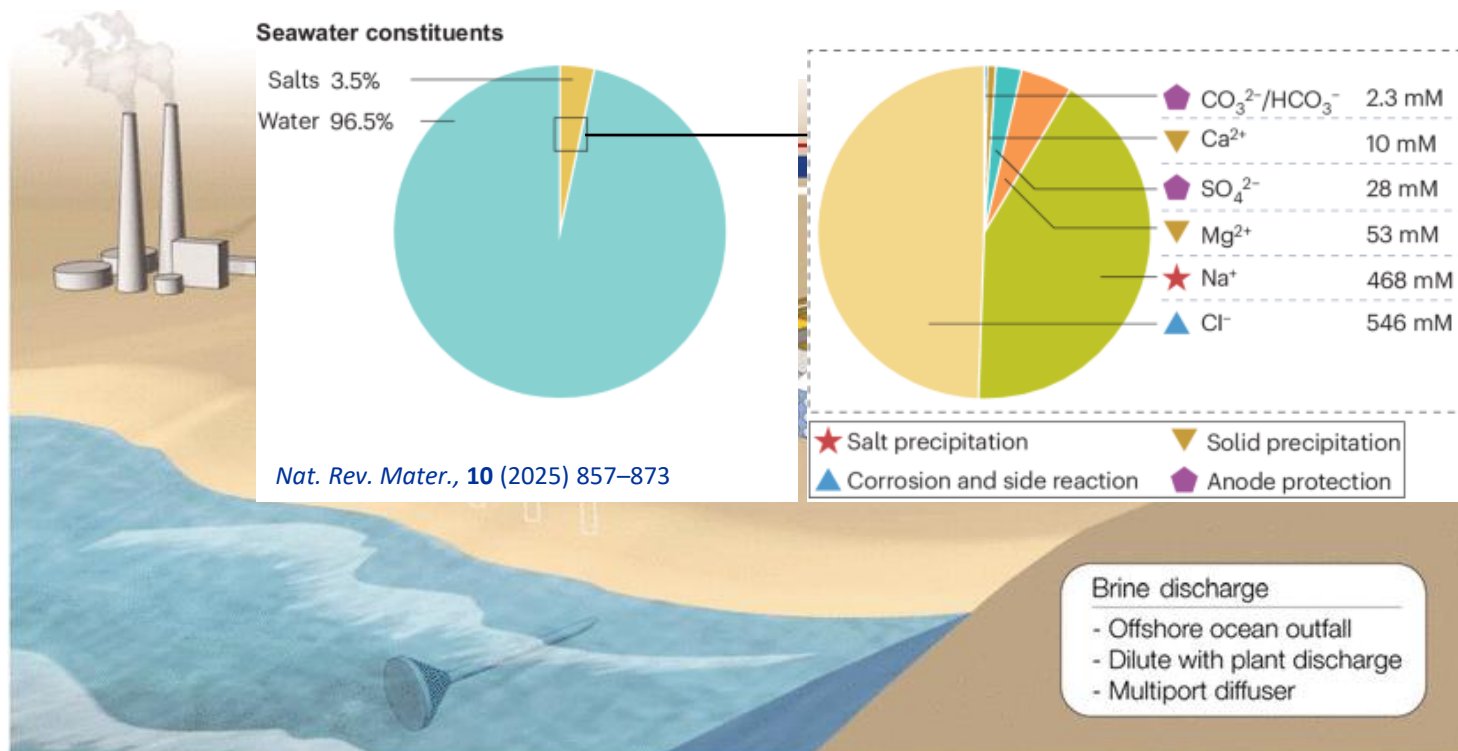
<Source: IRENA>

Electrolysis Hydrogen

□ Alkaline Water Electrolysis

Significant interest in direct seawater electrolysis (saline water electrolysis) using ALKs

- Annually 4.43 Gt (≈ 12 Mt per day) of H_2O is required to produce 492 Mt of H_2 (2050 target)
- Abundant seawater ($\sim 97\%$) vs. extremely scarce high-purity freshwater ($<1\%$ of total 13.9 Gt)



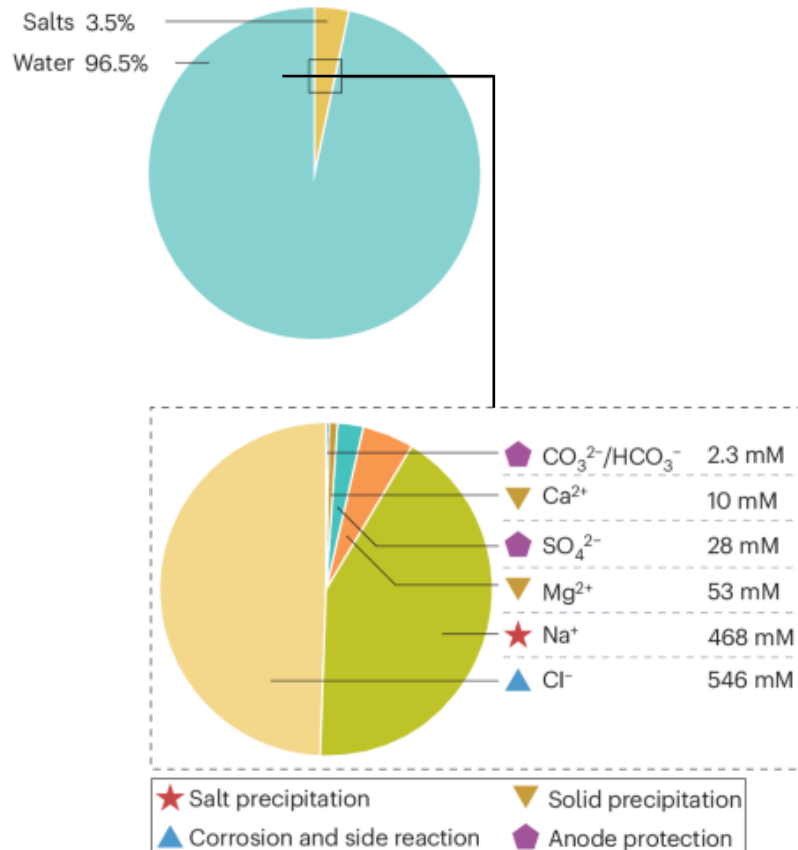
Science., **333**(6043) (2011) 712–717

Electrolysis Hydrogen

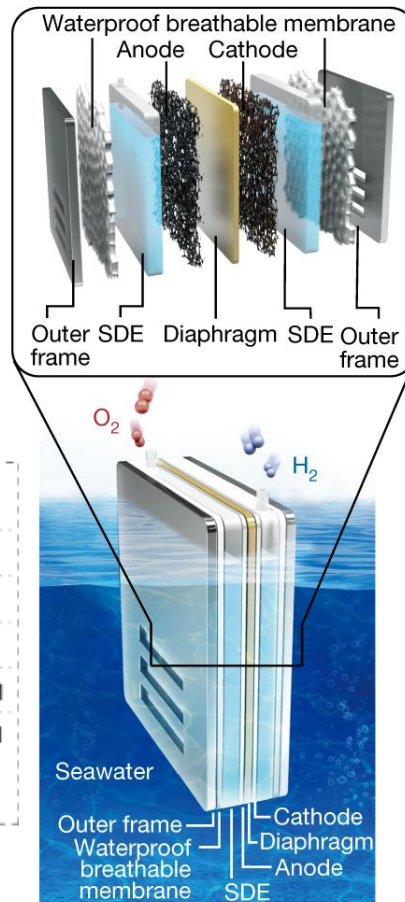
Alkaline Water Electrolysis

Significant interest in direct seawater electrolysis (saline water electrolysis) using ALKs

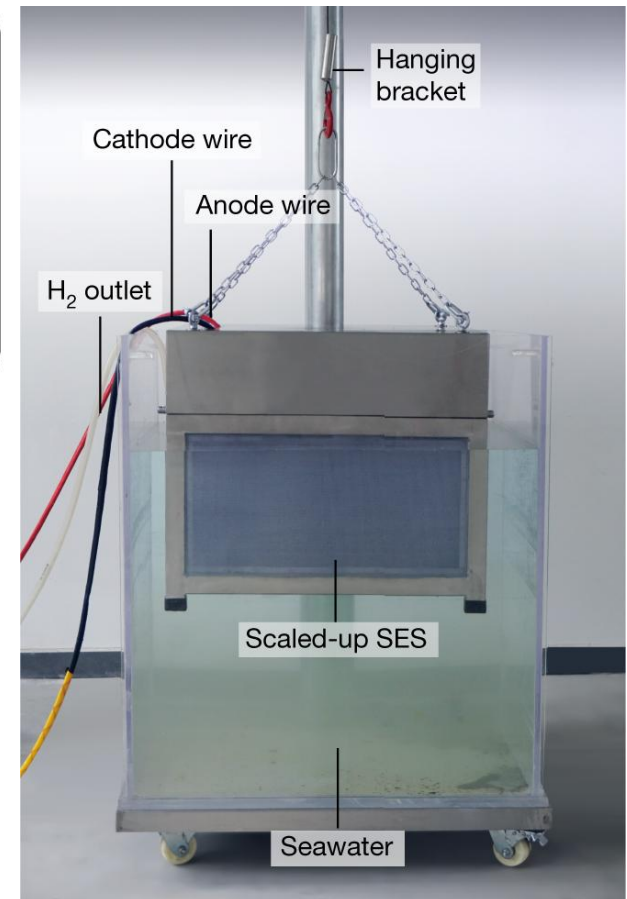
Seawater constituents



Nat. Rev. Mater., **10** (2025) 857–873



Nature, **612** (2022) 673–678

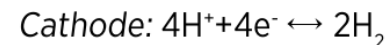
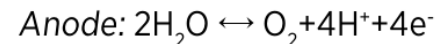
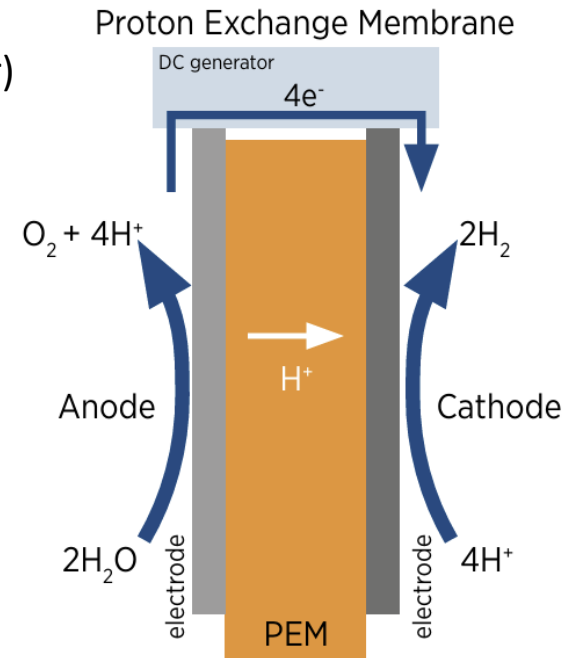
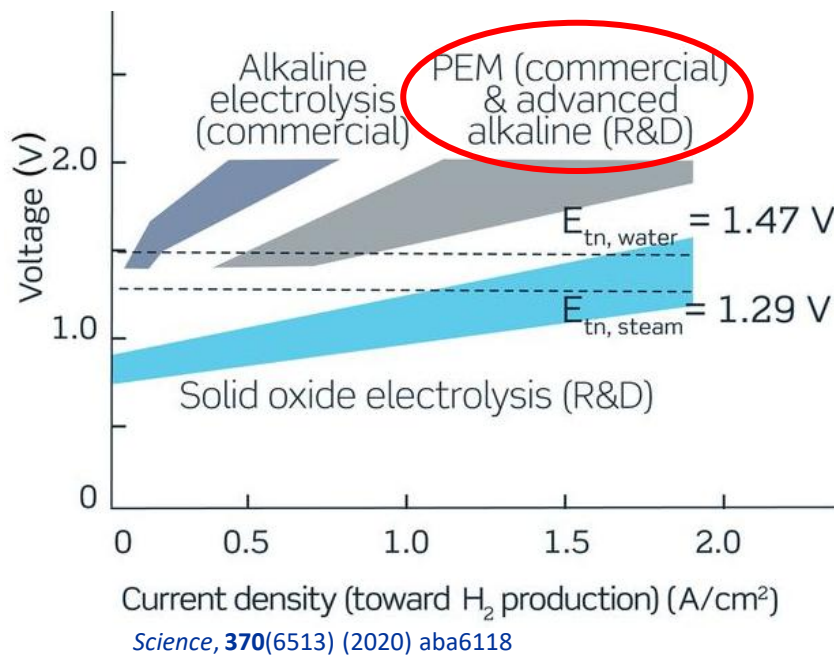


Electrolysis Hydrogen

□ Proton Exchange Membrane Water Electrolysis

Uses a *solid polymer electrolyte* to split pure H_2O into H_2 and O_2 using electricity

- A highly reliable, mature electrolysis technology with a technology readiness level (TRL) of 9
- Polymer membrane serving for ion exchange and stable electrolyte
- Simple, noncorrosive cell structure
- Highly dependent on a noble metal catalyst (e.g., Ir)



<Source: IRENA>

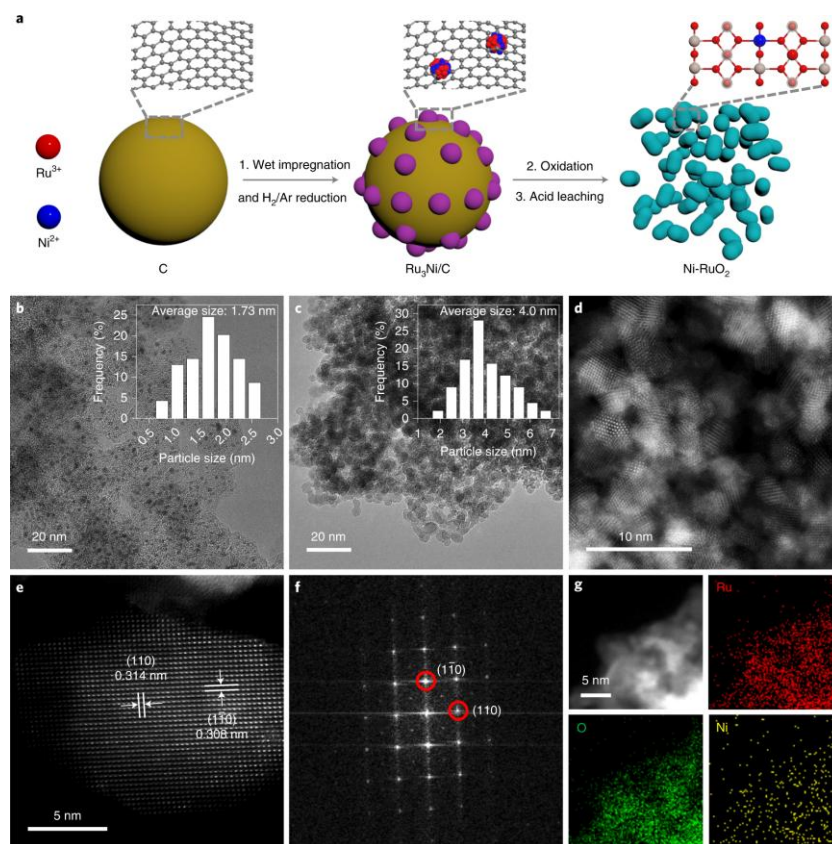
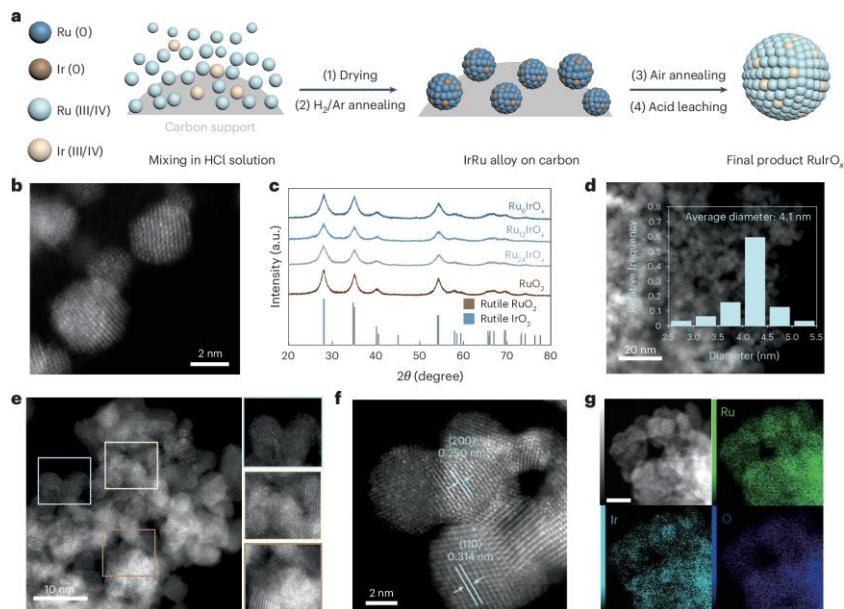
Electrolysis Hydrogen

□ Proton Exchange Membrane Water Electrolysis

Significant interest in reducing dependence on expensive *iridium catalysts*

Table Prices of noble metals as of 2025 (\$ kg⁻¹)

Ir	144,000	Ru	10,400
Au	75,430	Rh	147,000
Pt	27,800	Pd	49,500

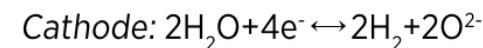
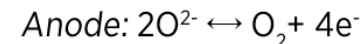
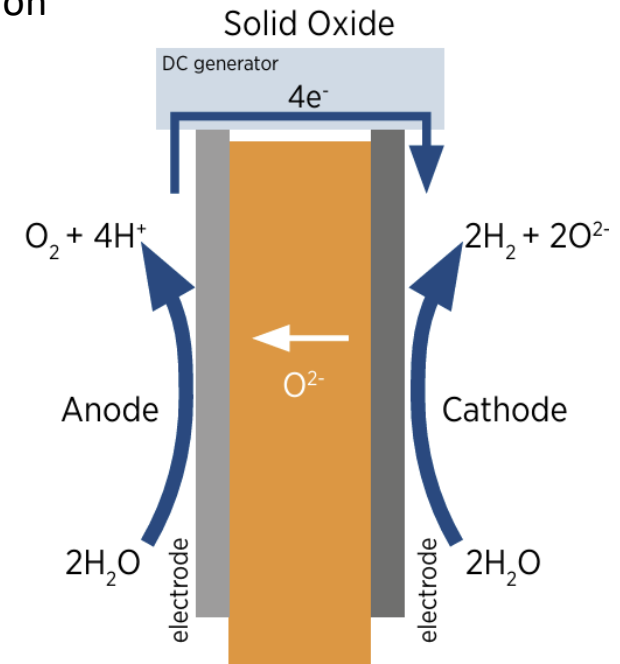
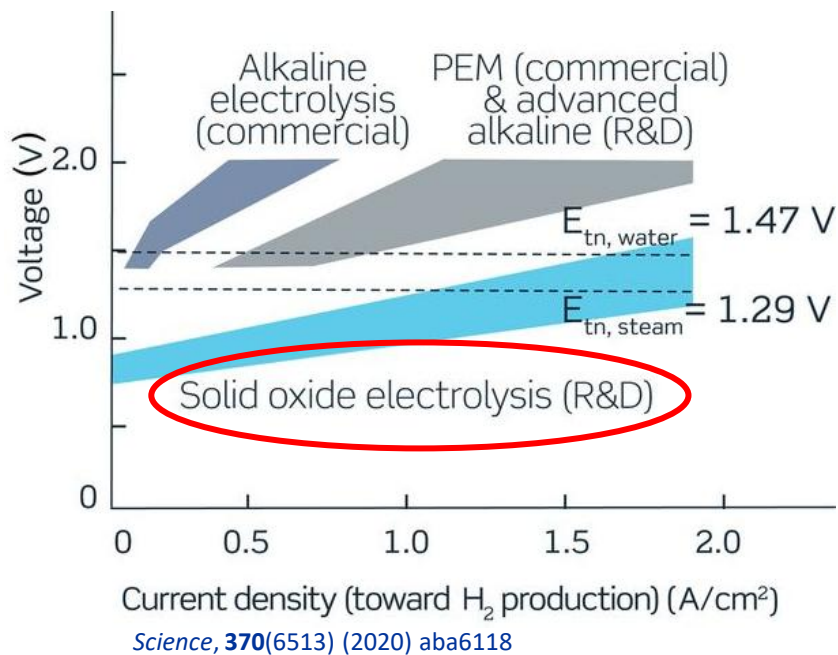


Electrolysis Hydrogen

❑ Solid Oxide Electrolysis

Uses a *solid oxide, or ceramic, electrolyte* to produce H_2 gas (and/or CO) and O_2

- Highly efficient device due to high-temperature operation ($>600\text{ }^{\circ}\text{C}$)
- Enables producing high-purity hydrogen
- Low competitiveness due to massive H_2 consumption

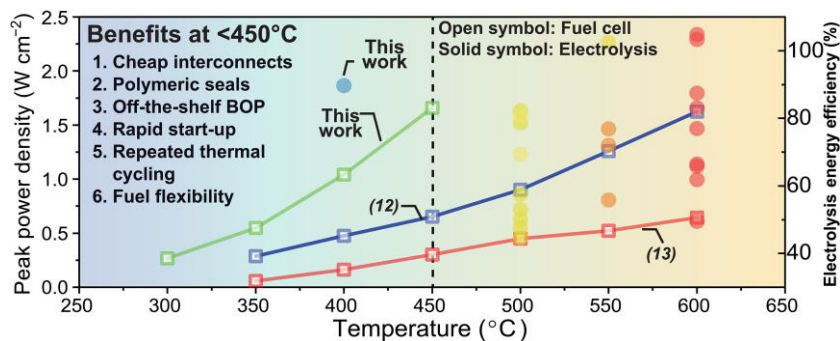


<Source: IRENA>

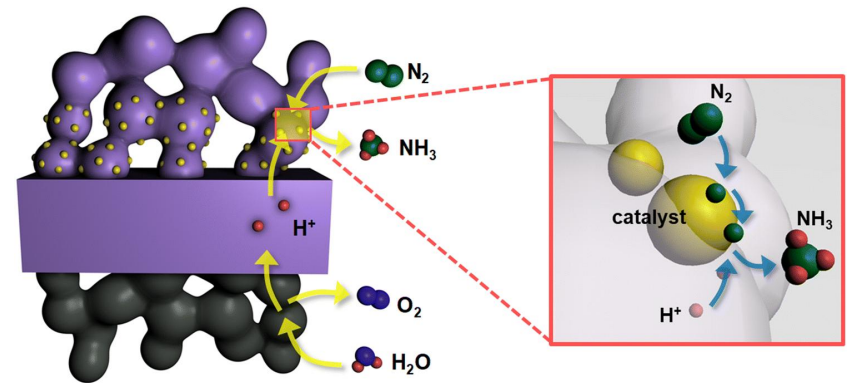
Electrolysis Hydrogen

□ Solid Oxide Electrolysis

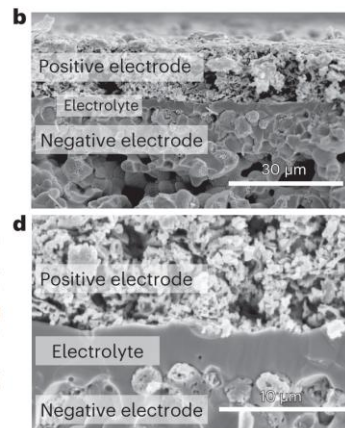
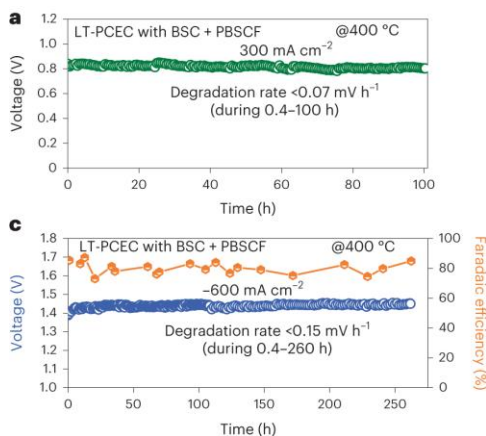
Significant focus on *lowering operating temp.* and *producing value-added chemicals* (CH_4 , NH_3 , etc.)



Sci. Adv., **11**(2) (2025) adq2507

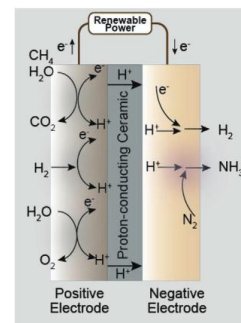


J. Korean Ceram. Soc., **57** (2020) 480–494

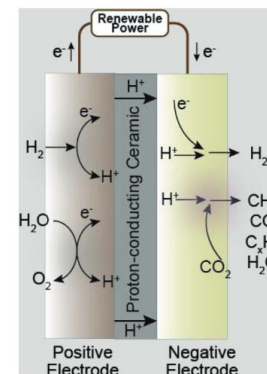


Nat. Energy, **8** (2023) 1145–1157

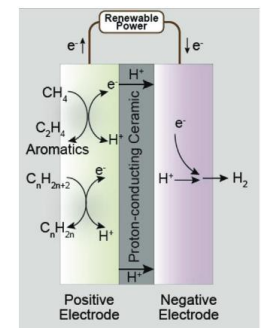
NH_3 synthesis



CO_2 reduction



C_mH_n conversion



Adv. Sci., **10** (2023) 2206478

Electrolysis Hydrogen

□ Comparison of Electrolysis Technology

	ALK	PEM	SO
Operating temp. (°C)	70–90	50–80	700–850
Operating pressure (bar)	1–30	<70	1
Electrolyte	5–7 mol L ⁻¹ KOH	PFSA membrane	Y₂O₃-stabilized ZrO₂ (YSZ)
Separator	ZrO ₂ stabilized with PPS mesh	Solid electrolyte (above)	Solid electrolyte (above)
Oxygen electrode	Ni-coated perforated stainless steel	IrO ₂	Perovskite oxide (e.g., LSCF, LSM)
Hydrogen electrode	(above)	Pt/C	Ni–YSZ
Porous transport layer (Anode)	Ni mesh	Pt-coated sintered porous Ti	Ni mesh
Porous transport layer (Cathode)	Ni mesh	Sintered porous Ti or C cloth	–
Bipolar plate (Anode)	Ni-coated stainless steel	Pt-coated Ti	–
Bipolar plate (Cathode)	Ni-coated stainless steel	Au-coated Ti	Co-coated stainless steel
Frames and Sealing	PSU, PTFE, EPDM	PTFE, PSU, ETFE	Ceramic glass

<Source: IRENA>

Electrolysis Hydrogen

□ Comparison of Electrolysis Technology (Today vs. 2050)

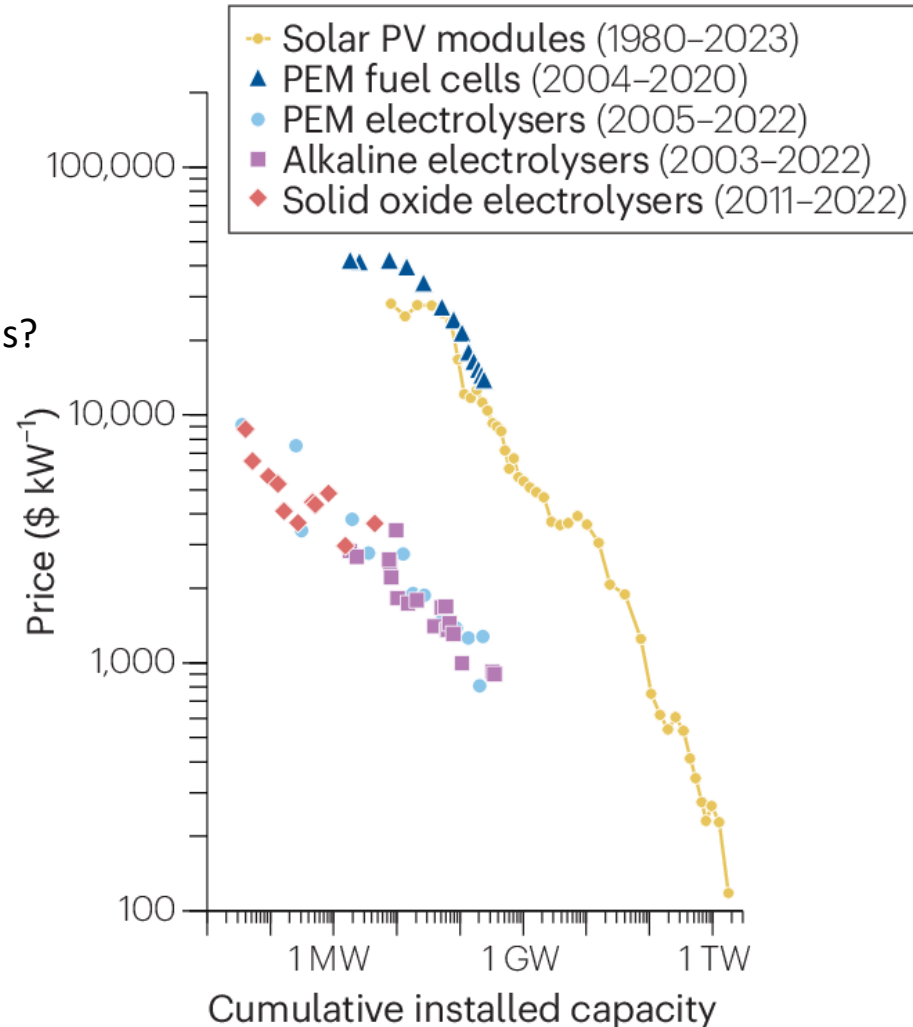
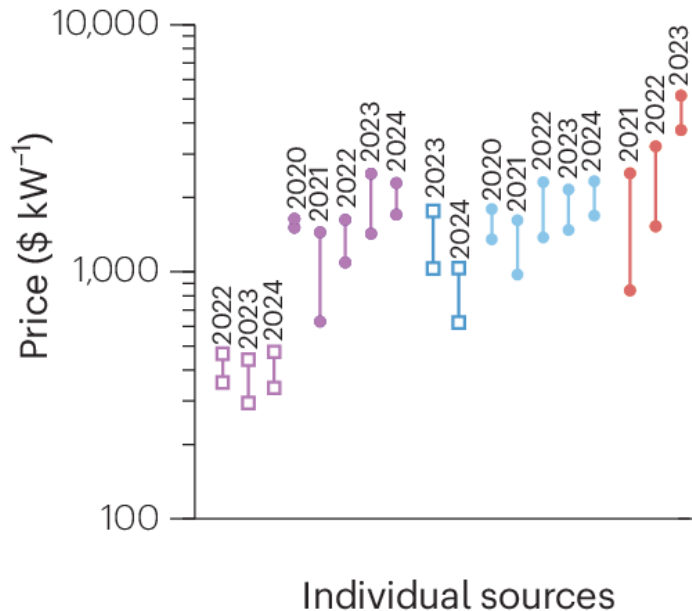
	2020			2050		
	ALK	PEM	SOEC	ALK	PEM	SOEC
Nominal current density (A cm^{-2})	0.2–0.8	1–2	0.3–1	>2	4–6	>2
Voltage range (V)	1.4–3.0	1.4–2.5	1.0–1.5	<1.7	<1.7	<1.48
Operating temp. ($^{\circ}\text{C}$)	70–90	50–80	700–850	>90	80	<600
Cell pressure (bar)	<30	<30	1	>70	>70	>20
H ₂ purity (%)	>99.9	>99.9	99.9	>99.9999	Same	>99.9999
Voltage eff. (LHV) (%)	50–68	50–68	75–85	70	>80	85
Electrical eff. (stack) (kWh kg^{-1})	47–66	47–66	35–50	<42	<42	<35
Electrical eff. (system) (kWh kg^{-1})	50–78	50–83	40–50	<45	<45	<40
Lifetime (system) (kh)	60	50–80	<20	100	100–120	80
Stack unit size (MW)	1	1	0.005	10	10	0.2
Electrode area (cm^2)	>10000	1500	200	30000	>10000	500
Startup (to nominal load) (min)	<50	<20	>600	<30	< 5	<300
CAPEX (stack) (>1 MW) ($\text{\$ kW}^{-1}$)	270	400	>2000	<100	<100	<200
CAPEX (system) (>10 MW) ($\text{\$ kW}^{-1}$)	500–1000	700–1400	N/A	<200	<200	<300

Electrolysis Hydrogen

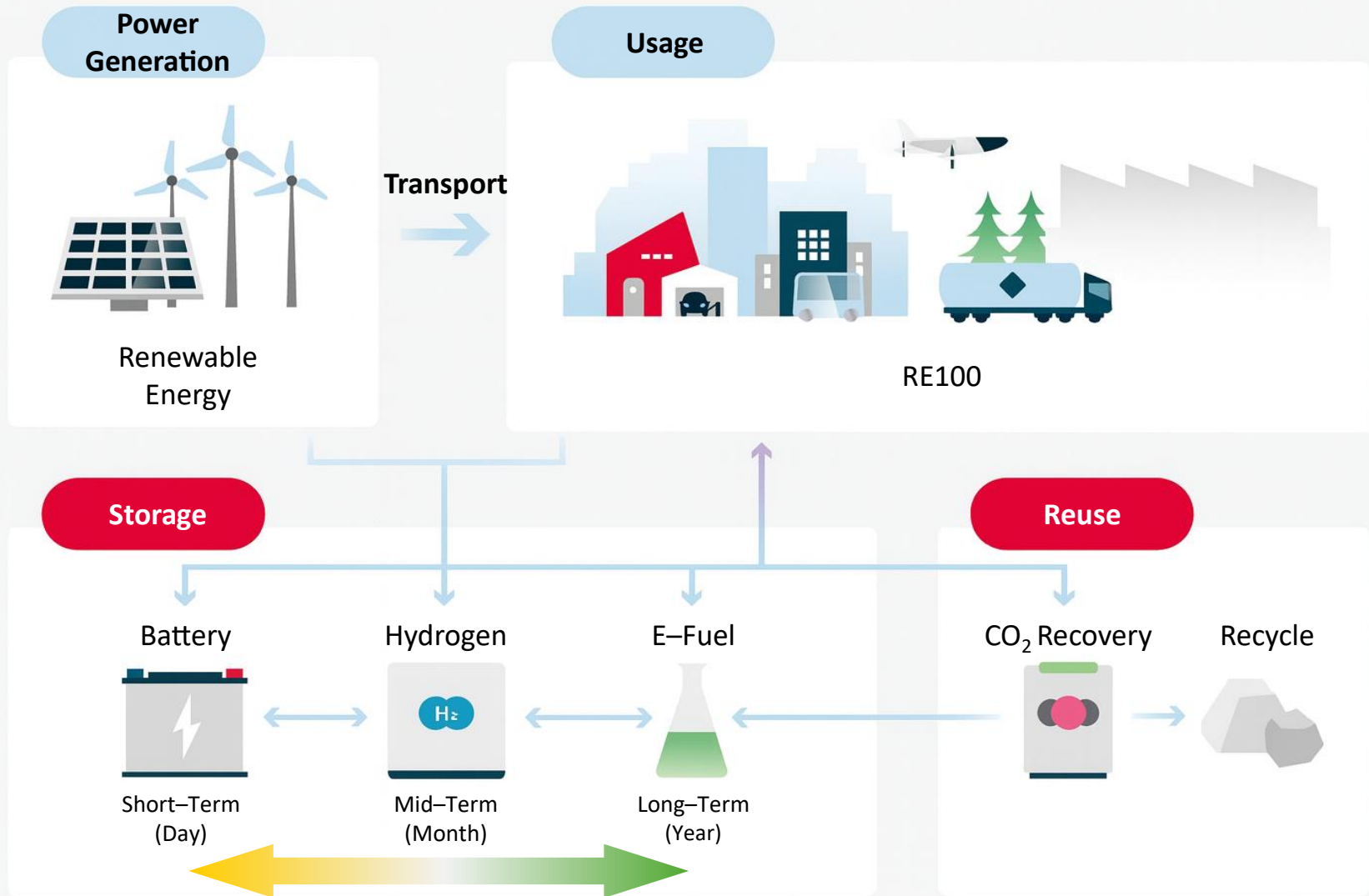
Levelized Cost of Hydrogen (LCOH) & Capital Cost (CAPEX)



Global inflation? Skyrocketing raw material prices?



Surplus RE Hydrogen Utilization



Key Technological Objectives

❑ Produce, store/transport, and use

Establishing, standardizing, and commercializing secure, low-cost technologies across these *three primary pillars* is essential for introducing a full hydrogen society

PRODUCE

- Low cost
- Large scale
- Carbon-free production

STORE & TRANSPORT

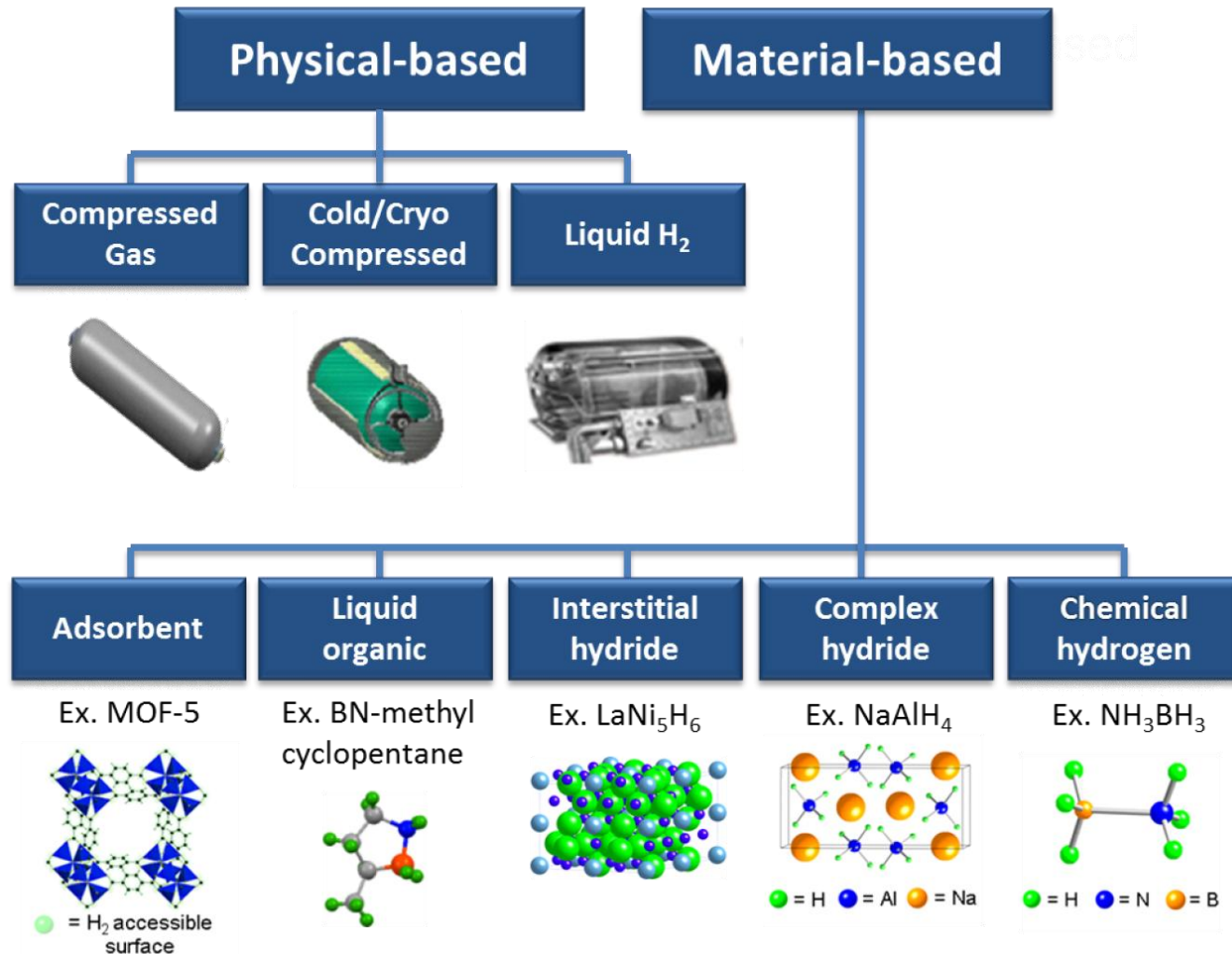
- High density
- Loss free
- Safe contaminant

USE

- High efficiency
- Clean energy
- Safe energy

Hydrogen Storage Technology

How is hydrogen stored?



<Source: US Department of Energy>

Hydrogen Storage Technology

□ Energy Density of H₂

H₂ has high mass energy density, but *low volumetric energy density* at STP (0 °C, 1 atm)

- **Mass energy density**

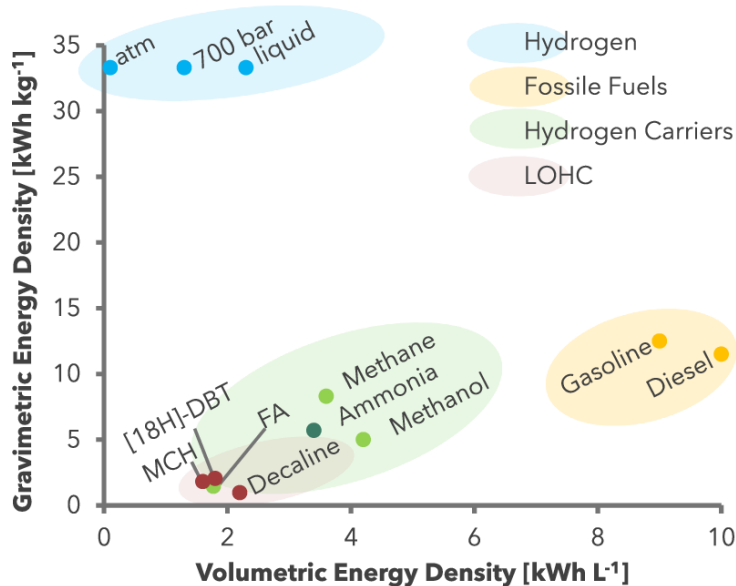
$$241.8 \text{ kJ mol}^{-1} \div 0.002016 \text{ kg mol}^{-1} = 119.9 \text{ MJ kg}^{-1} \text{ (cf. Gasoline: } \sim 44 \text{ MJ L}^{-1}\text{)}$$

- **Volumetric energy density**

$$241.8 \text{ kJ mol}^{-1} \div 22.4 \text{ L mol}^{-1} = 0.01079 \text{ MJ L}^{-1} \text{ (cf. Gasoline: } \sim 32 \text{ MJ L}^{-1}\text{)}$$

Table Volume and density of compressed H₂

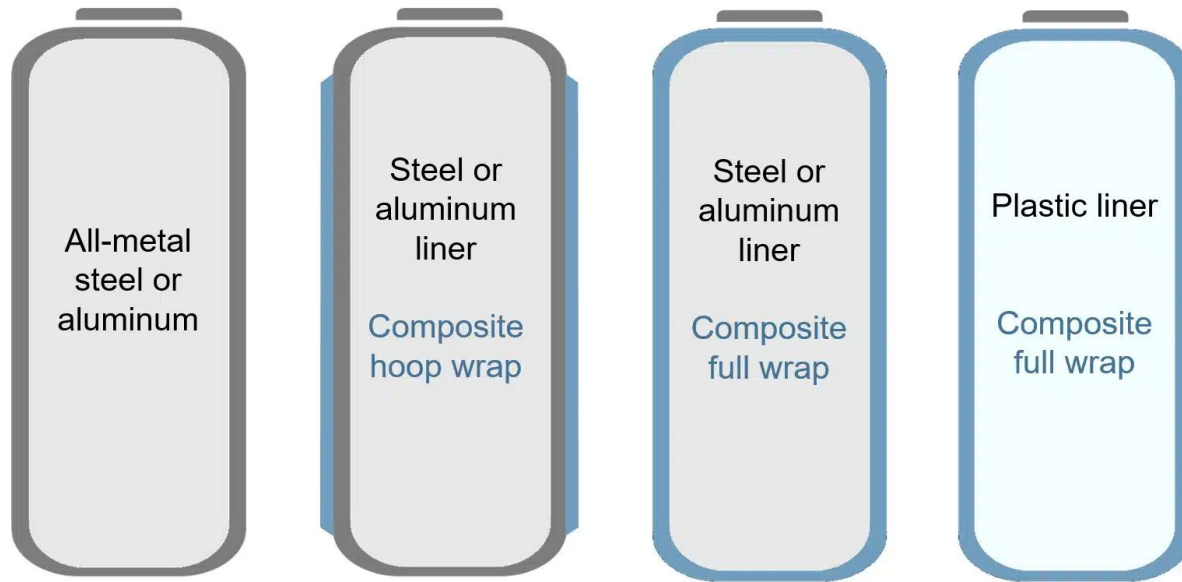
Pressure (bar)	Normalized volume (%)	Density (kg m ⁻³)	Remarks
1.0	100	0.0887	Ambient
100	1.1	8.32	Industry
200	0.57	15.6	Tube trailer
350	0.35	24.9	Heavy-duty
700	0.21	41.1	FCV
900	0.18	47.9	H ₂ station



Physical-Based H₂ Storage

❑ Compressed Gaseous H₂ (CGH₂)

Compressed hydrogen in tanks at 350 and 700 bar is used for mobile hydrogen storage in FCVs



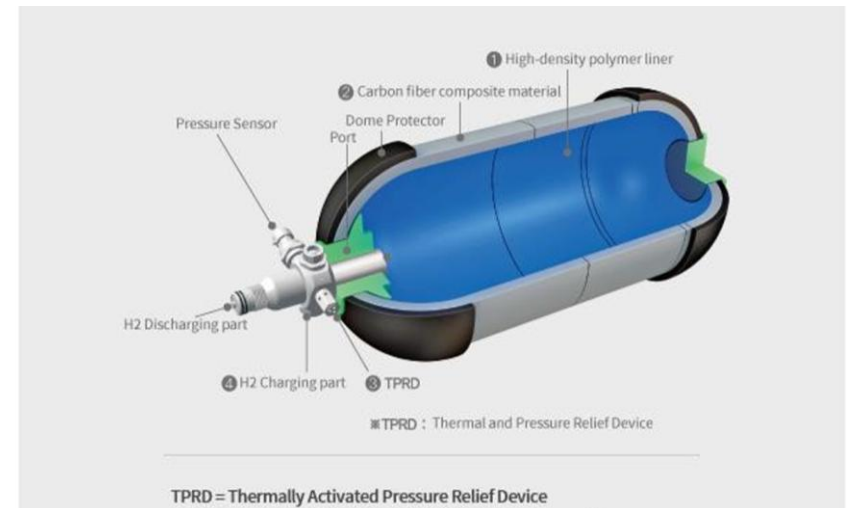
	Type I	Type II	Type III	Type IV
Pressure	200 bar	<300 bar	350–700 bar	350–700 bar
Relative Weight	3.3	2.7	1.4	1.0
Relative Capacity	30%	35%	75%	100%
Cost	Lowest	Low	High	Slightly high *\$15 kWh ⁻¹ (= \$500 kg _{H₂} ⁻¹) (2015)
Stability	Low	Low	High	High

Physical-Based H₂ Storage

❑ Hydrogen Truck (Tube Trailer)

Trucks that haul gaseous hydrogen, which is typically compressed to 180 bar or higher into long cylinders stacked on the trucks.

- **Energy for compression**
Typically, $\sim 2\text{--}4.4 \text{ kWh kg}_{\text{H}_2}^{-1}$
- **Cost for compression**
Depends on distance, electricity price, and pressure



수소 튜브트레일러란?		타입1(Type1) 수소 튜브트레일러	타입4(Type4) 수소 튜브트레일러
수소 튜브트레일러란?		· 특정차에 기체 수소 저장 용기가 장착된 차량을 의미 · 기체 수소 저장 용기에 따라 구분	
기체 수소 저장 용기	재질	· 타입1 저장용기 사용 · 용기 전체가 금속재질 라이너(내측)로 구성	· 타입4 저장용기 사용 · 플라스틱과 같은 비금속 라이너(내측)에 탄소섬유 복합재료로 용기 전체를 보강한 형태
	입력	· 주로 200bar 사용하나, · 150~300bar까지 저장 가능	· 제조사별 상이하나, · 350~700bar까지 저장 가능
수송 용량		· 최대 수송용량: 340kg · 수소 튜브트레일러 중량 대비 수소 저장량이 작음	· 최대 수송용량: 634kg~1,000kg(개발 중) · 기존 대비 운송비 절감

<Source: KPMG>

Eur. J. Energy Res., 5(2) (2025)

Physical-Based H₂ Storage

□ Hydrogen Pipeline

Used to connect the point of hydrogen production or delivery of hydrogen with the point of demand

- **As of 2024**

USA – 2,600 km of low pressure hydrogen pipelines

Europe – 1,600 km of low pressure hydrogen pipelines

<Source: EPRI, World Economic Forum>

- **Price**

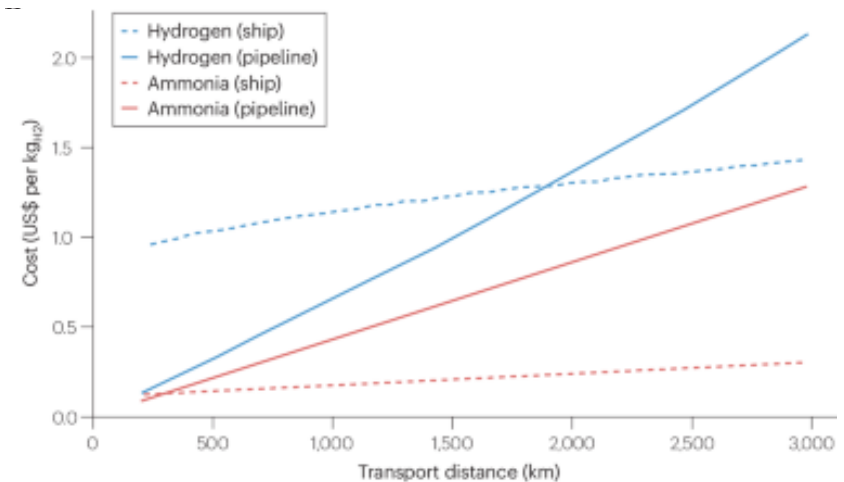
\$~0.5–1.0 kg_{H₂}⁻¹ per 1,000 km

- **Weakness**

Requires massive initial investment and dedicated special piping due to *high leakage risks*, *metal embrittlement*, and *a wide flammability range*.



<Source: ABC News>



Nat. Comm., 15 (2024) 7268

Physical-Based H₂ Storage

❑ Liquified H₂ (LH₂)

The liquid state of hydrogen, which is commonly used as a liquid rocket fuel for rocketry application

*NASA, US Air Force, etc.

- **Low boiling point**

–252.9 °C (=20.3 K) at atmospheric pressure (1 atm)

Liquefaction consumes more than 30% of the energy content of the hydrogen and is expensive.

**Theoretically, H₂ requires ~3.4 kWh kg⁻¹ to liquify

***Existing liquefaction facilities use 10–13 kWh kg⁻¹ (LHV 33 kWh kg⁻¹)

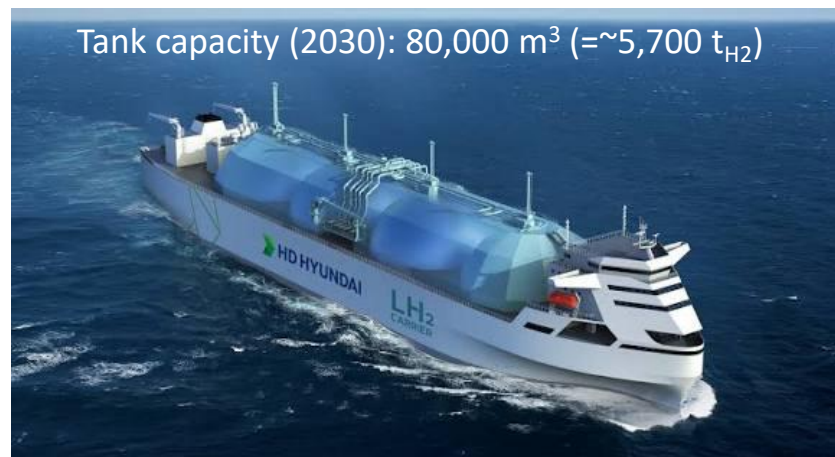
- **Higher density**

Its density is 70.9 g L⁻¹, which is ~790 and ~1.73 times higher than those of atmospheric H₂ (0.090 g L⁻¹) and compressed H₂ at 700 bar (41.1 g L⁻¹).

But it is still light (relative density = 0.07), compared to gasoline (0.74) and LNG (0.45).



<Source: Iwatani Corp.>



<Source: HD Hyundai>

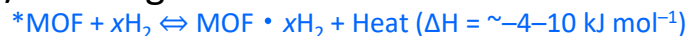
Material-Based H₂ Storage

☐ Metal–Oxide Framework (MOF)

A method of storing hydrogen molecules via physical adsorption within ultra-porous 3D crystalline structures composed of metal ions and organic linkers

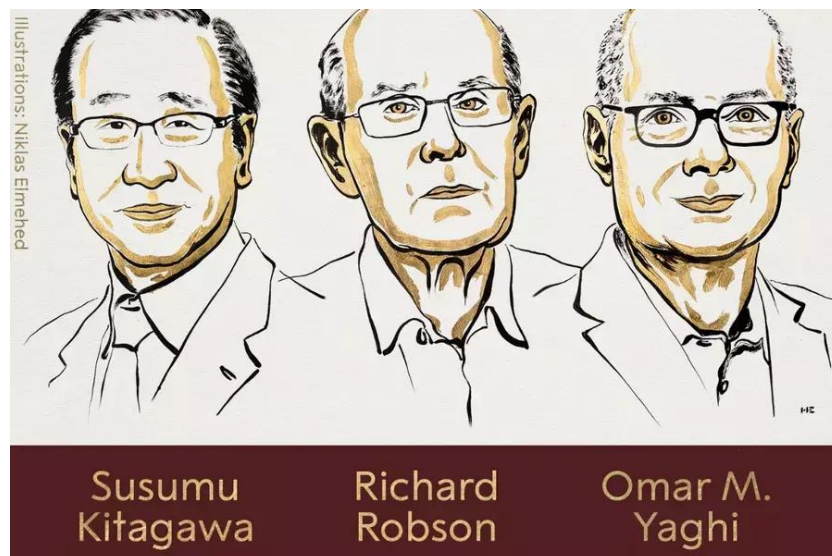
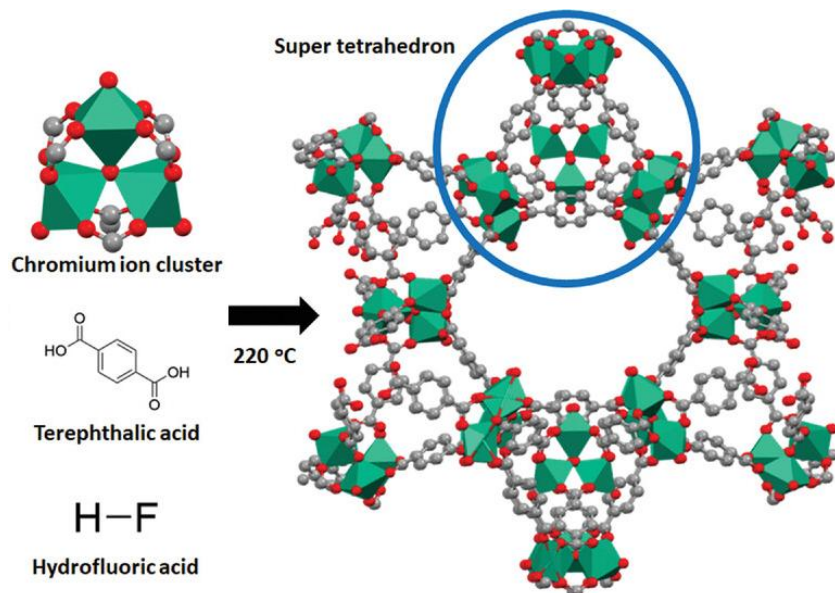
- **Fast Kinetics & High Capacity**

High gravimetric capacity (7–14 wt%) at moderate pressures (30–100 bar) with fully reversible, rapid charge/discharge



- **Cryogenic Dependency**

Low storage capacity at RT due to poor adsorption efficiency, requiring cryogenic cooling ($\sim 77 \text{ K}$)



Material-Based H₂ Storage

☐ Metal Hydride

A method of storing in the form of compounds with light metals (such as Li, B, Mg, and Al) or carbon (such as activated carbon and carbon nanotubes)

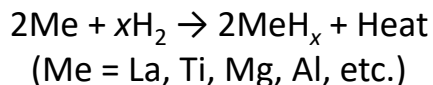
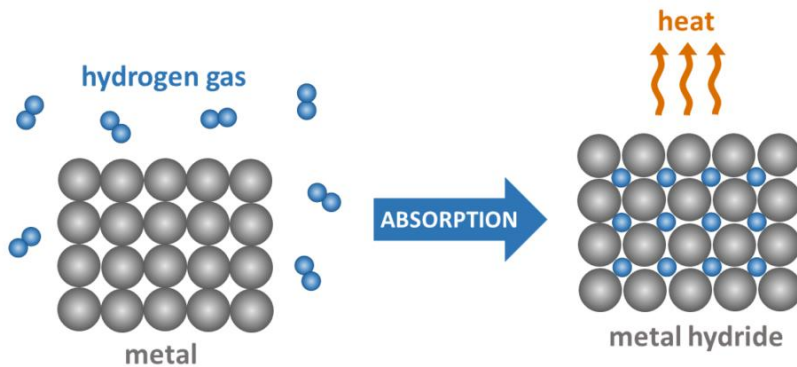
- **Safety**

Operates at low pressures (often <10–50 bar) with high volumetric capacity (~100 kg m⁻³) and low explosion risks compared to high-pressure gas tanks

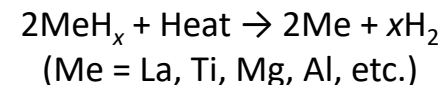
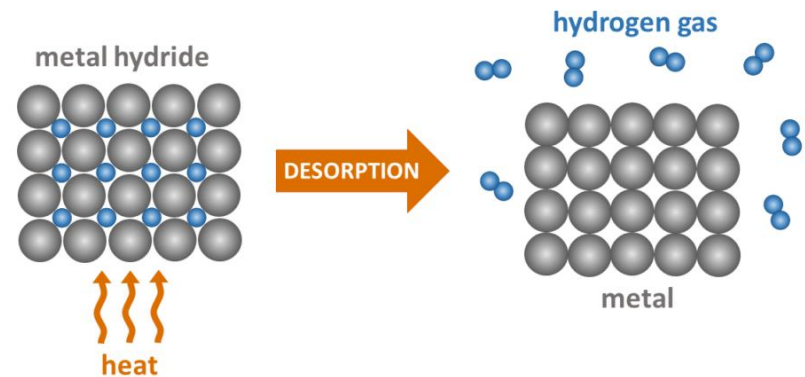
- **Heavy Weight**

Low gravimetric capacity (~1–2 wt%), making it heavy for mobile transportation

Exothermic Reaction



Endothermic Reaction



Material-Based H₂ Storage

❑ Liquid Organic Hydrogen Carrier (LOHC)

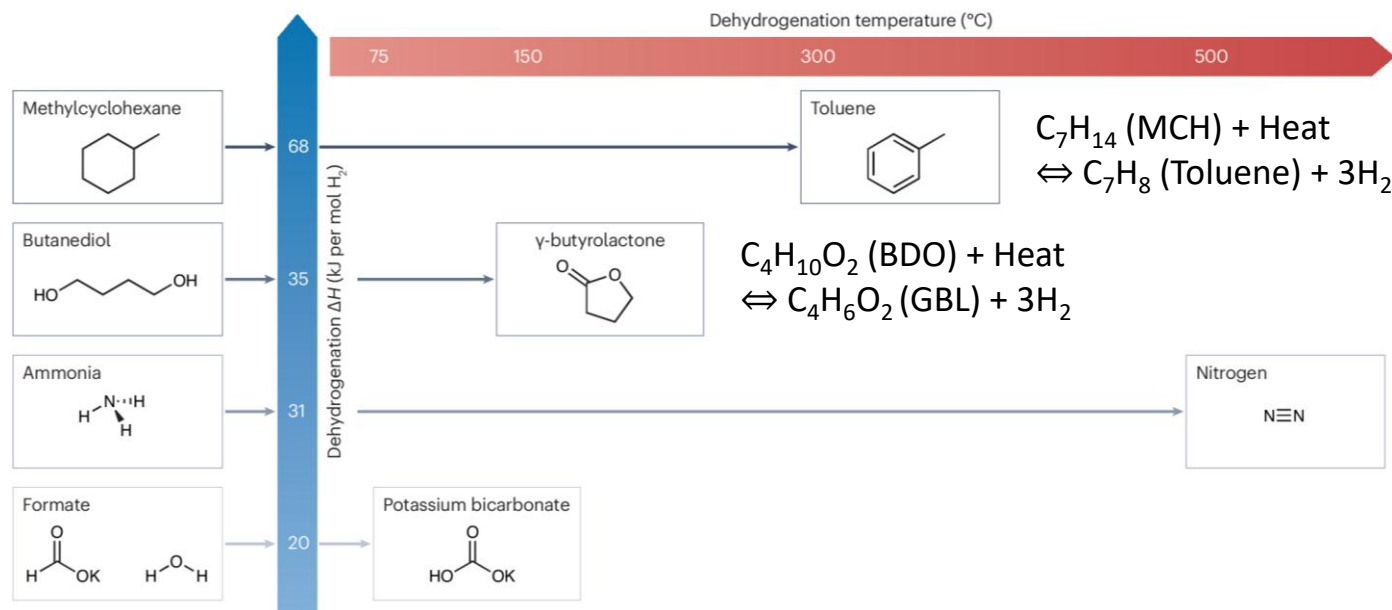
A technology that allows hydrogen to be chemically bonded to a liquid carrier (e.g., organic oil), making it safer and more efficient to handle

- **Uses Petroleum Infrastructure**

Liquid at ambient condition; 100% compatible with existing tankers and oil pipelines

- **High Heat Demand**

Requires high-temperature heat input (~250–300 °C) and noble catalysts for hydrogen release



Material-Based H₂ Storage

□ Ammonia (NH₃)

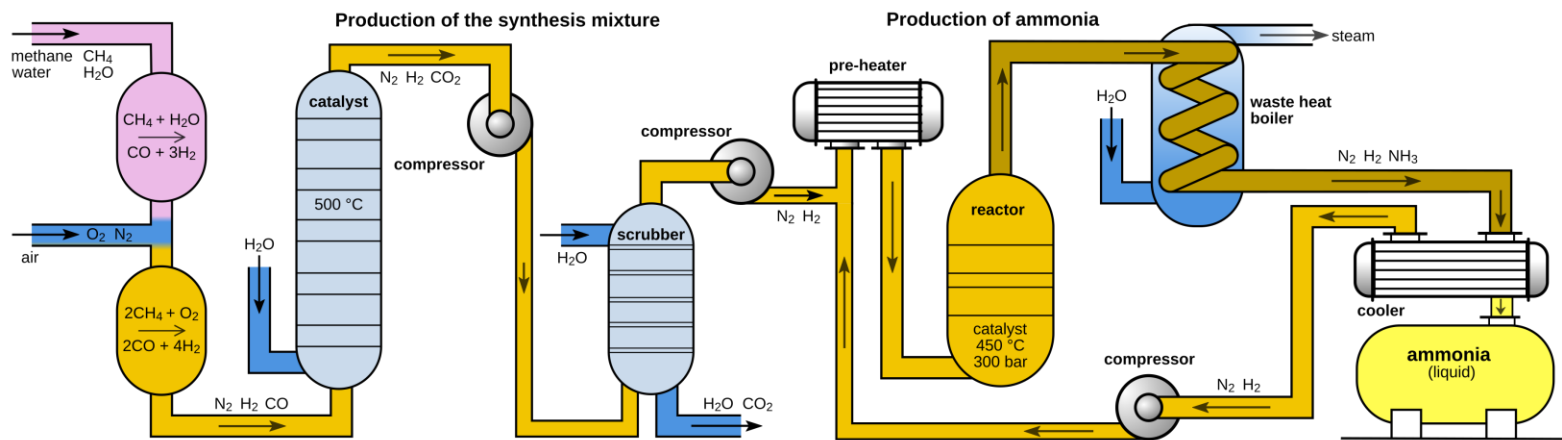
A technology that allows hydrogen to be chemically bonded to a liquid carrier (e.g., organic oil), making it safer and more efficient to handle

- **Highest Hydrogen Density**

Delivers high gravimetric capacity (17.7 wt%) with zero carbon emissions while utilizing mature global ammonia shipping infrastructure

- **Toxicity & High Cracking Energy Demand**

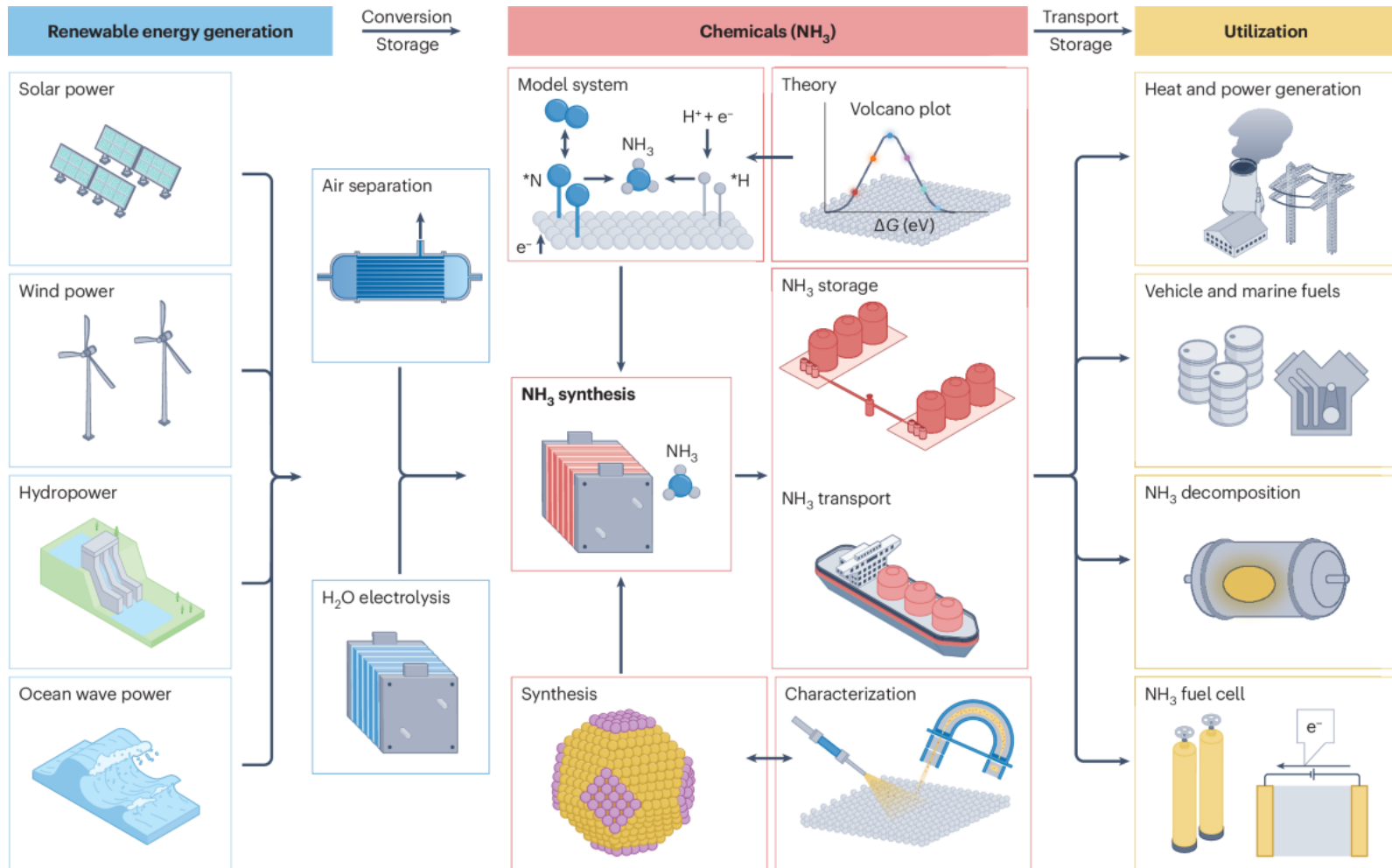
Requires strict safety protocols due to toxicity and high thermal energy input (500 °C) for endothermic cracking into pure H₂



<Source: https://en.wikipedia.org/wiki/Haber_process>

Material-Based H₂ Storage

Ammonia (NH₃)



Hydrogen Storage Technology

□ Technology Comparison

Technology	State	Volumetric density	Gravimetric density	Key advantage	Challenges
CGH ₂	Gas (350–700 bar, Ambient)	Low (~40 kg m ⁻³)	4.5–5.7 wt%	Fast refueling Mature tech	High pressure Vessel cost
LH ₂	Liquid (1–10 bar, –253 °C)	High (~70 kg m ⁻³)	11–20 wt%	Low volume	High energy Boil-off gas
MOF	Solid (30–100 bar, –77 °C)	High (~50–60 kg m ⁻³)	7.0–10 wt%	Fast kinetics Low pressure	Cryogenic temp. Low RT capacity
Metal Hydride	Solid (10–50 bar, Ambient)	Very high (~100 kg m ⁻³)	1.0–1.5 wt%	Low pressure Safety	Heavy mass Heat mgmt.
LOHC	Liquid (Ambient Pressure/Temp)	Medium (~60 kg m ⁻³)	6.0–6.2 wt%	Uses existing infrastructure	High release heat
Ammonia	Liquid (8.5 bar or –33 °C)	Very high (~120 kg m ⁻³)	17.7 wt%	High capacity Carbon-free	Toxicity Cracking cost

Key Technological Objectives

❑ Produce, store/transport, and use

Establishing, standardizing, and commercializing secure, low-cost technologies across these *three primary pillars* is essential for introducing a full hydrogen society

PRODUCE

- Low cost
- Large scale
- Carbon-free production

STORE & TRANSPORT

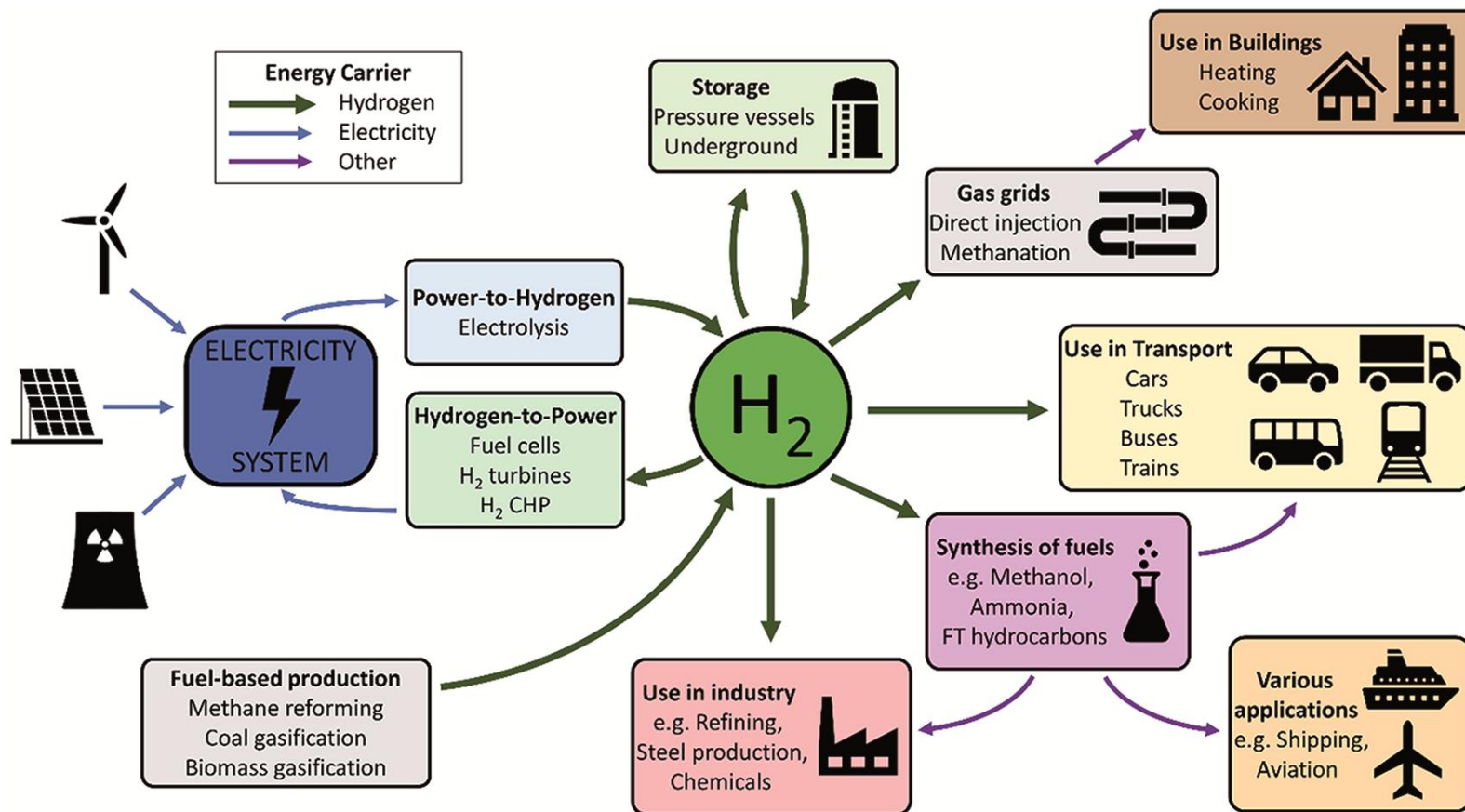
- High density
- Loss free
- Safe contaminant

USE

- High efficiency
- Clean energy
- Safe energy

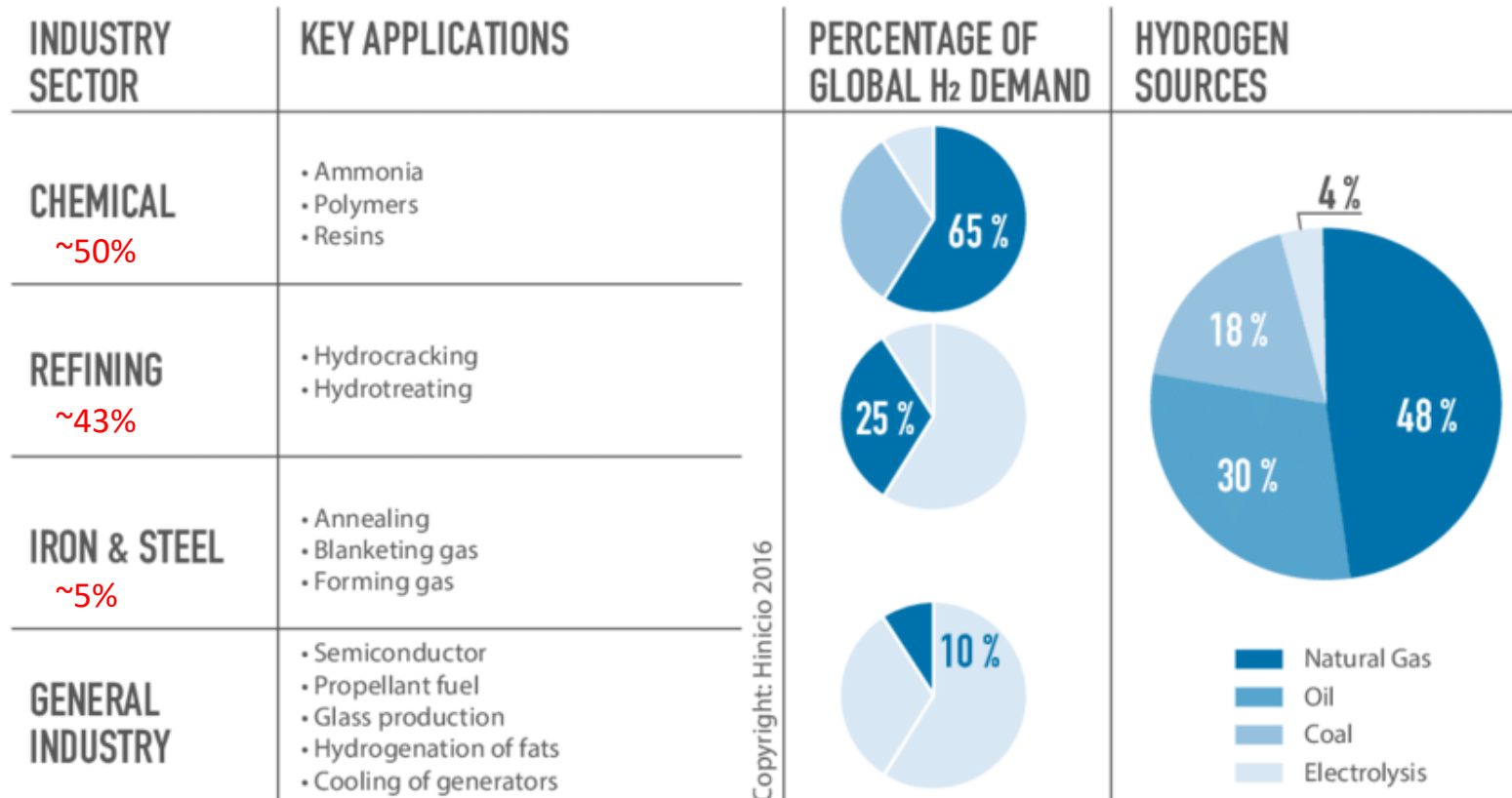
Hydrogen Use

Global Hydrogen Use by Sector



Hydrogen Use

Global Hydrogen Use by Sector



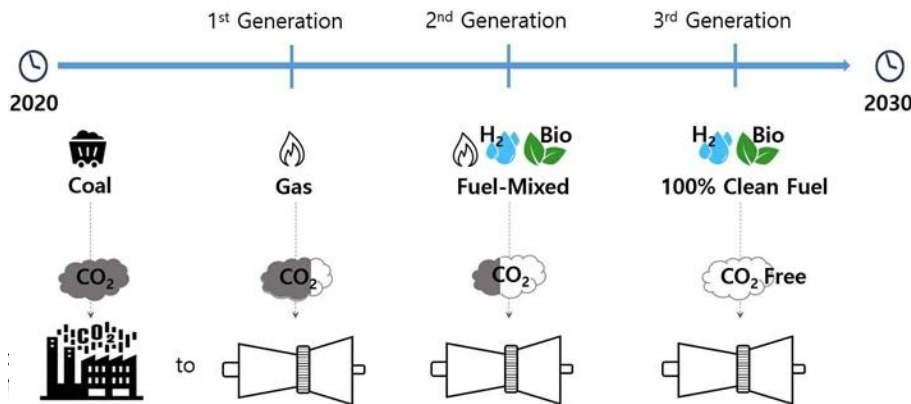
<Source: IRENA (2018)>

- Of the 100 Mt of H₂ produced in 2021, 43% was used in oil refining and 57% in industry, principally in the manufacture of NH₃ for fertilizers, and CH₃OH

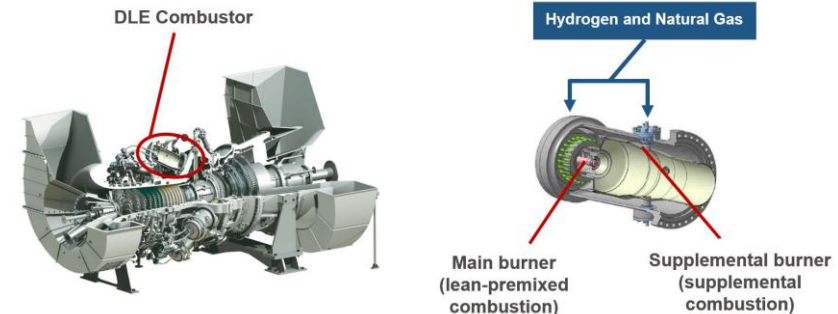
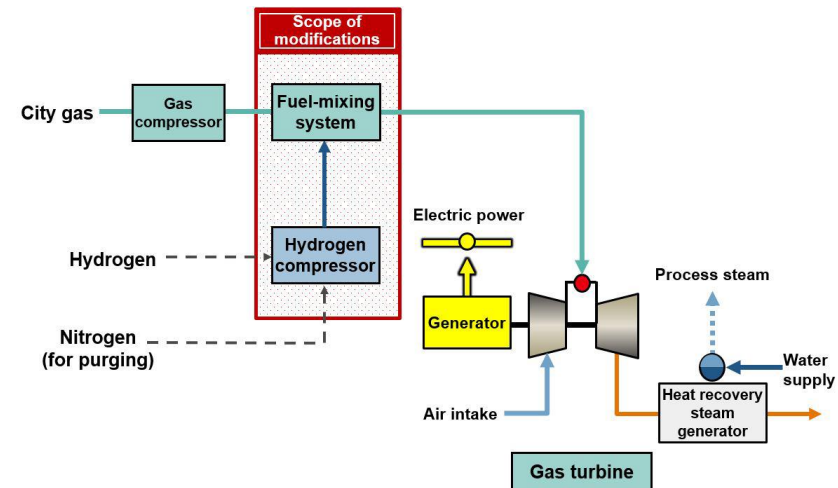
Hydrogen Use

□ Hydrogen Gas Turbine (GT)

Unlike conventional GTs operated by LNG, hydrogen GTs emit zero CO₂ during power generation, as their combustion process releases only H₂O



Properties		Unit	Methane	Hydrogen
Formula		-	CH ₄	H ₂
Molecular weight		g/mol	16	2
Density		kg/nm ³	0.716	0.090
Heating value (LHV)	Weight	MJ/kg	50	120
	Volume	MJ/nm ³	35.8	10.8



<Source: https://global.kawasaki.com/en/corp/newsroom/news/detail/?f=20220802_0328>

Hydrogen Use

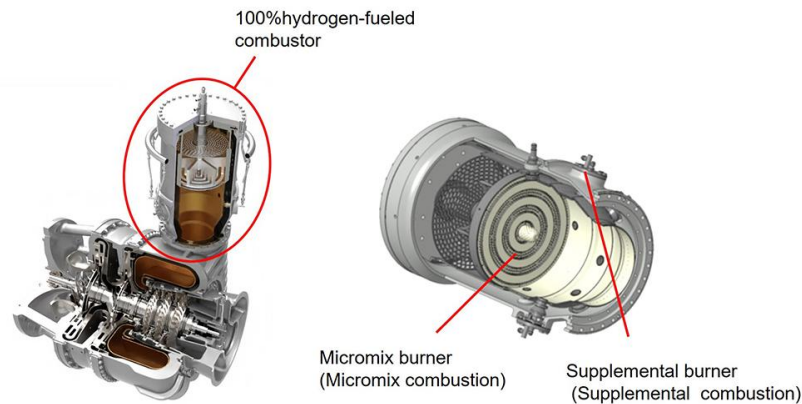


❑ Hydrogen Gas Turbine (GT)

Unlike conventional GTs operated by LNG, hydrogen GTs emit zero CO₂ during power generation, as their combustion process releases only H₂O

Kawasaki Launches World's First 1.8 MW Class, 100% Hydrogen-fueled, Dry-combustion Gas Turbine Cogeneration System

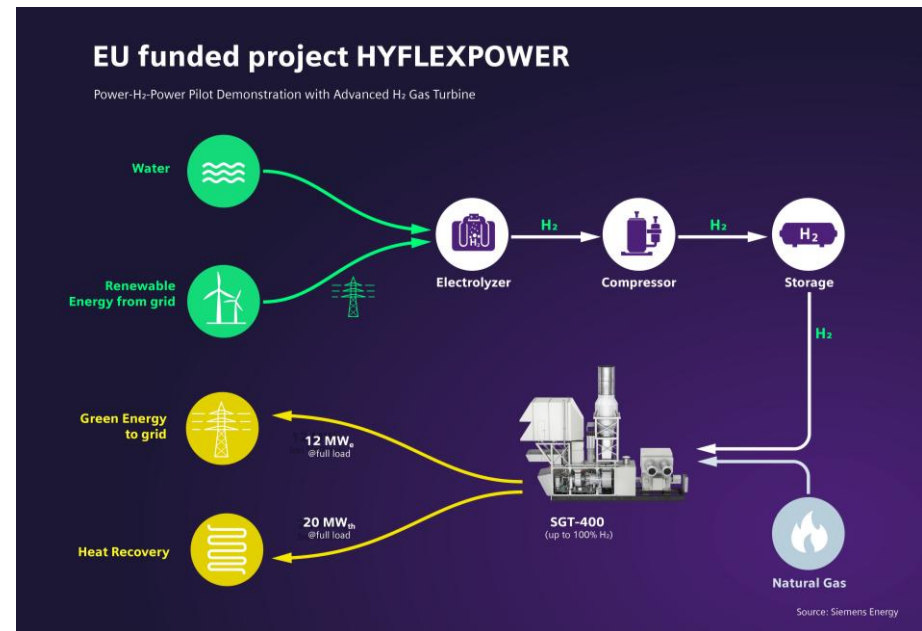
Sep. 05, 2023



<Source: https://global.kawasaki.com/en/corp/newsroom/news/detail/?f=20230905_2781>

Siemens Energy team demonstrates gas turbine running on 100% hydrogen

Oct 17, 2023, 11:12:02 AM | Article by [Plamena Tisheva](#)



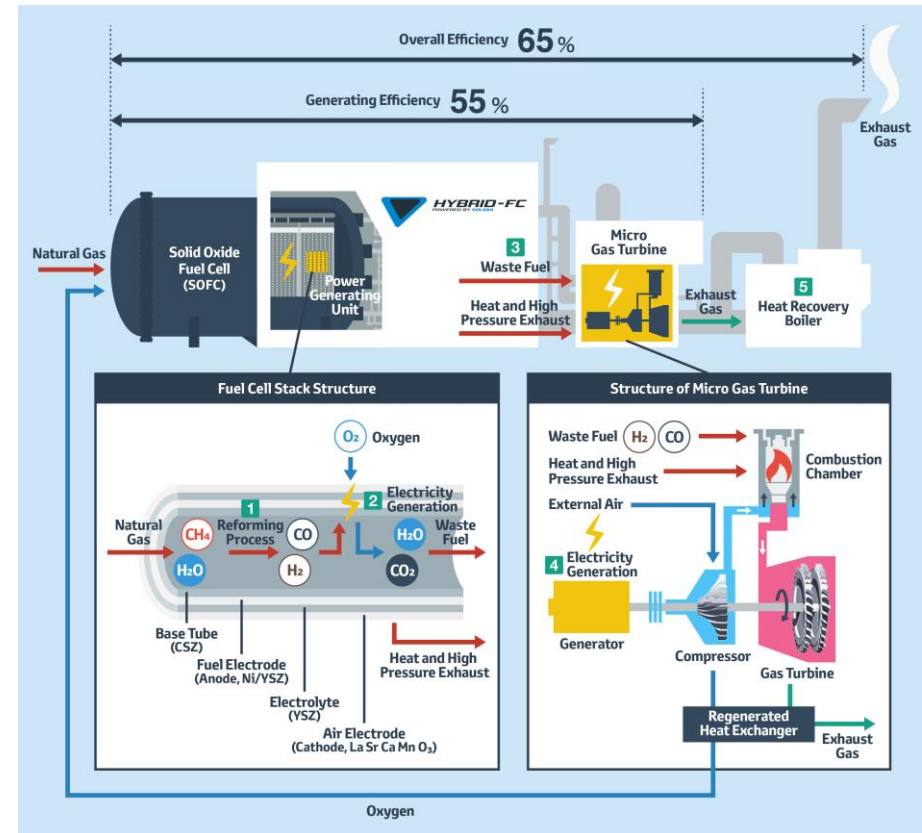
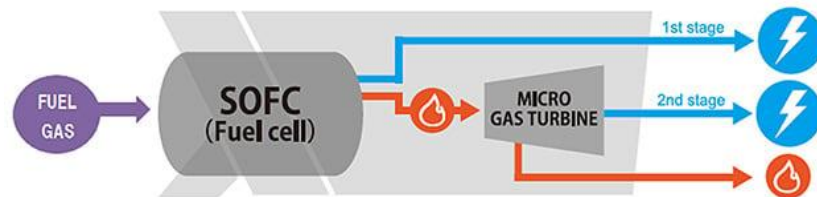
<Source: <https://www.siemens-energy.com/global/en/home/press-releases/hyflexpower-worlds-first-integrated-power-x-power-hydrogen-gas-turbine-demonstrator.html>>

Hydrogen Use

□ Gas Turbine × Fuel Cell

A synergistic integration of a high-temperature fuel cell (e.g., SOFC) with a micro gas turbine cycle

Toyota Starts Trial of a Hybrid Power Generation System Combining Fuel Cell Technology with Micro Gas Turbines at Motomachi Plant

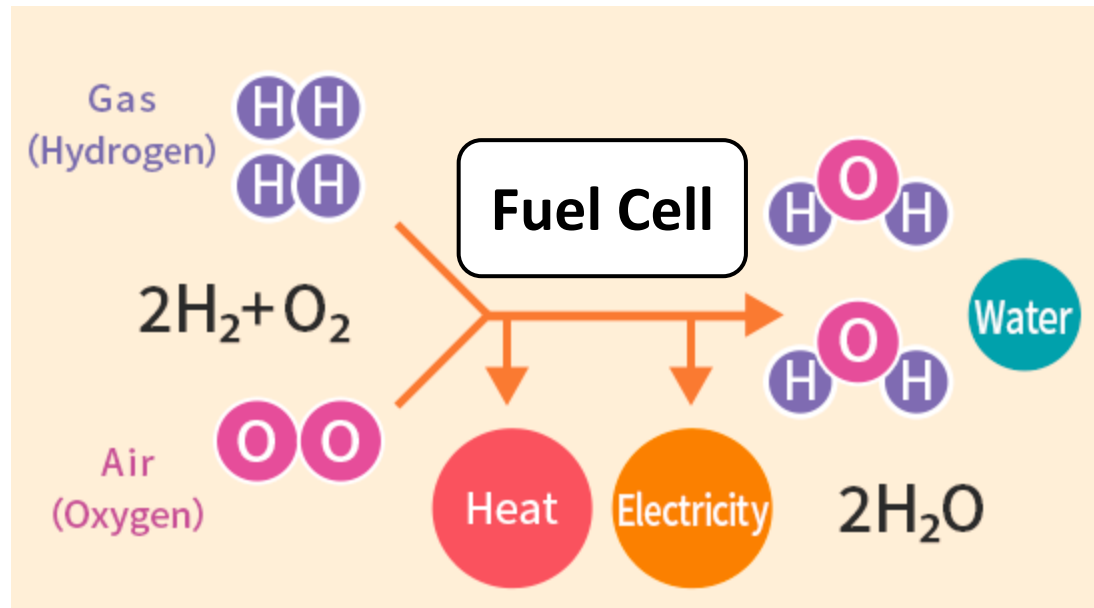


<Source: <https://media.toyota.ca/en/releases/2017/toyota-starts-trial-of-a-hybrid-power-generation-system-combining-fuel-cell-technology-with-micro-gas-turbines-at-motomachi-plant.html>>

Hydrogen Use

❑ Fuel Cell

An electrochemical energy conversion device that can directly convert chemical energy to electricity

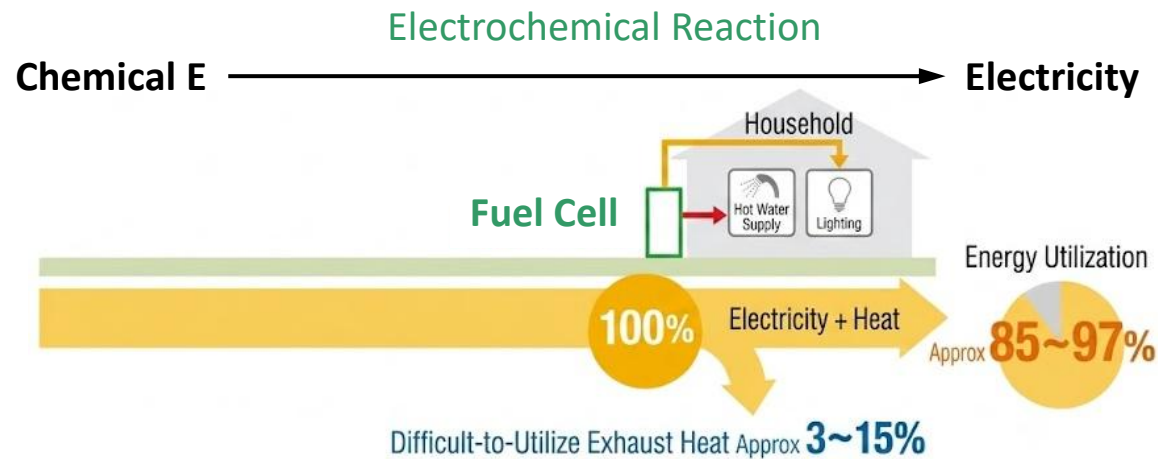
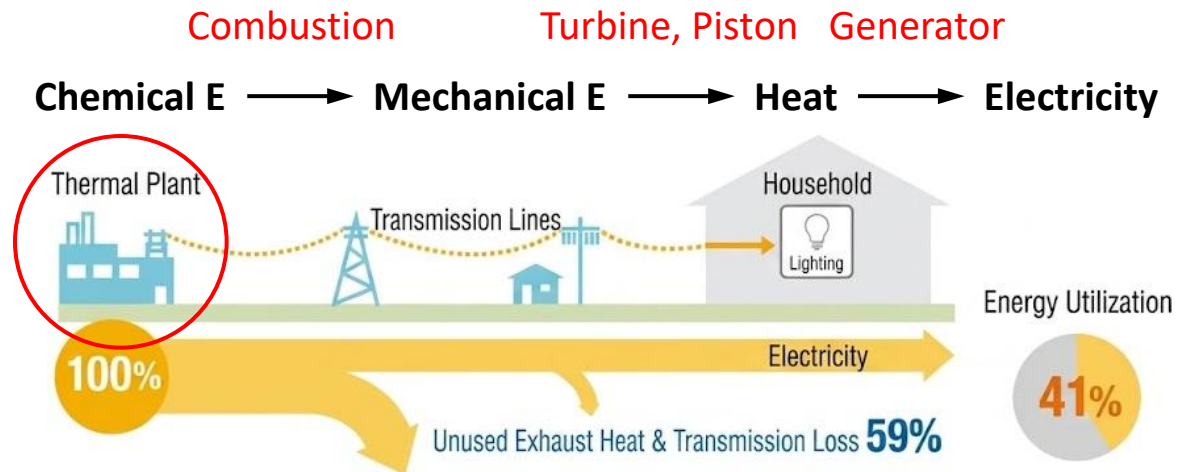


- Water and/or CO₂ are produced from fuels such as H₂, C_mH_n, CO, NH₃, etc. with O₂ as the oxidant
- Partial conversion of the chemical energy difference between reactants and products into electricity

Fuel Cell

❑ Thermal Plant vs. Fuel Cell

An electrochemical energy conversion device that can directly convert chemical energy to electricity



Fuel Cell



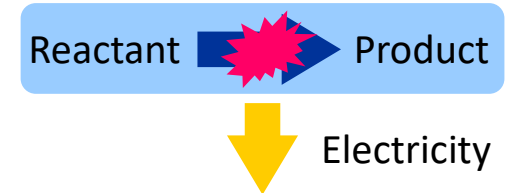
□ Characteristics of Fuel Cell

- **Primary Cell**

Stores reactants and products inside the container

Disposable (Single-use)

(e.g.) Zinc–carbon (Manganese) battery, Alkaline battery

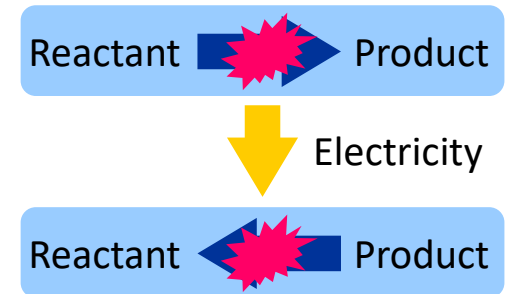


- **Secondary Cell**

Stores reactants and products inside the container

Reusable (Rechargeable battery)

(e.g.) Li-ion battery, Lead–acid battery, Ni–Cd battery, etc.

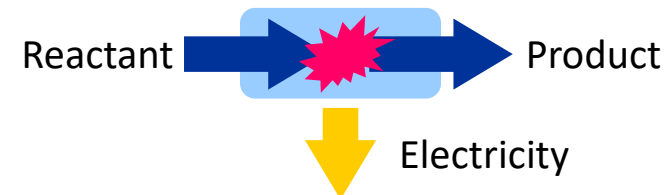


- **Fuel Cell**

Reactants are supplied from the outside

Products are exhausted to the outside

(e.g.) PEMFC, SOFC, etc.



Fuel Cell



❑ Characteristics of Fuel Cell

- **Continuous power generation is possible as long as reactants are supplied**
Differs from conventional batteries (primary and secondary cells)
- **Expected Applications**
 - Stationary power generation
 - Vehicles / Mobile equipment (Motive power, Auxiliary power supply)
 - Alternative to secondary batteries
(Charging → Refueling: Saves time)
 - (Higher fuel energy density leads to smaller and lighter systems)



Fuel Cell

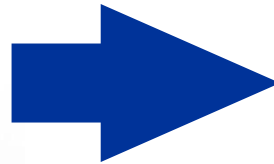
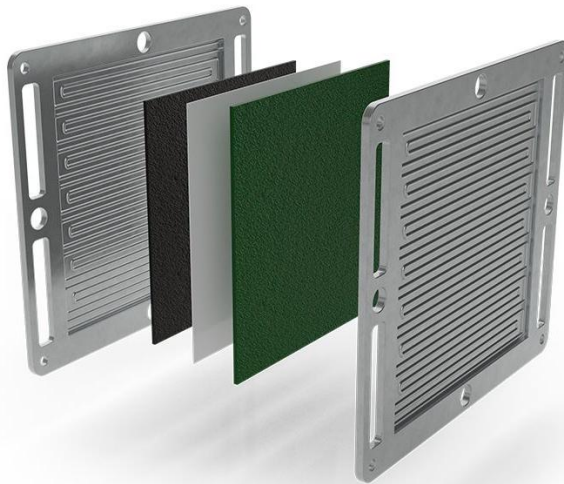


□ Fuel Cell Applications

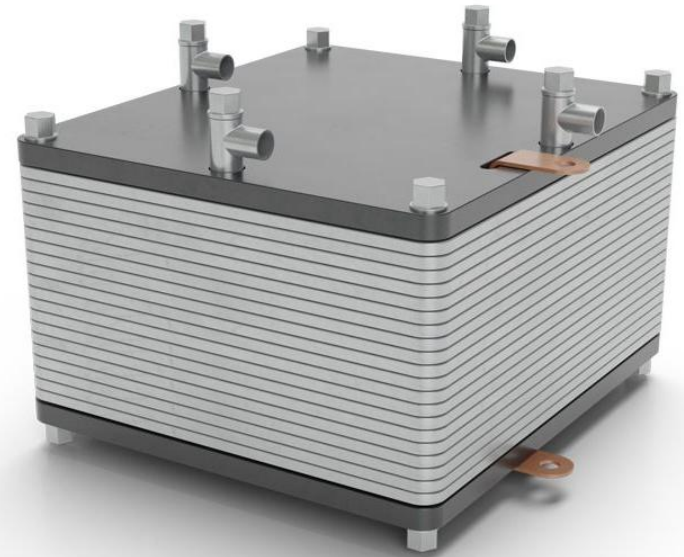
용량	고정 (발전소)	육상교통 (자동차, 열차)		수상교통 (선박)	공중교통 (비행기, 우주선)	
1~3 kW	가정용	전동기			드론	우주왕복선
20~100 kW	건물용	승용차		어선	0-2인 ^[14]	우주정거장
200~400 kW ^[15]		상용차	동차/트램	여객선/잠수함	3-5인 ^[16]	달 기지
1~4 MW	단지용		기관차	크루즈선	6-20인 ^[17]	
4~10 MW					비행기	
40~100 MW				소형 화물선/군함/잠수함 ^[18]		
100~1000 MW ^[19]	도시용			대형 화물선/군함/잠수함		

Fuel Cell

□ Single Cell & Stack



**Connected
in series
(e.g., 40-cell stack)**



Nominal current: 30 A

Rated voltage: $\sim 0.7\text{--}0.9$ V

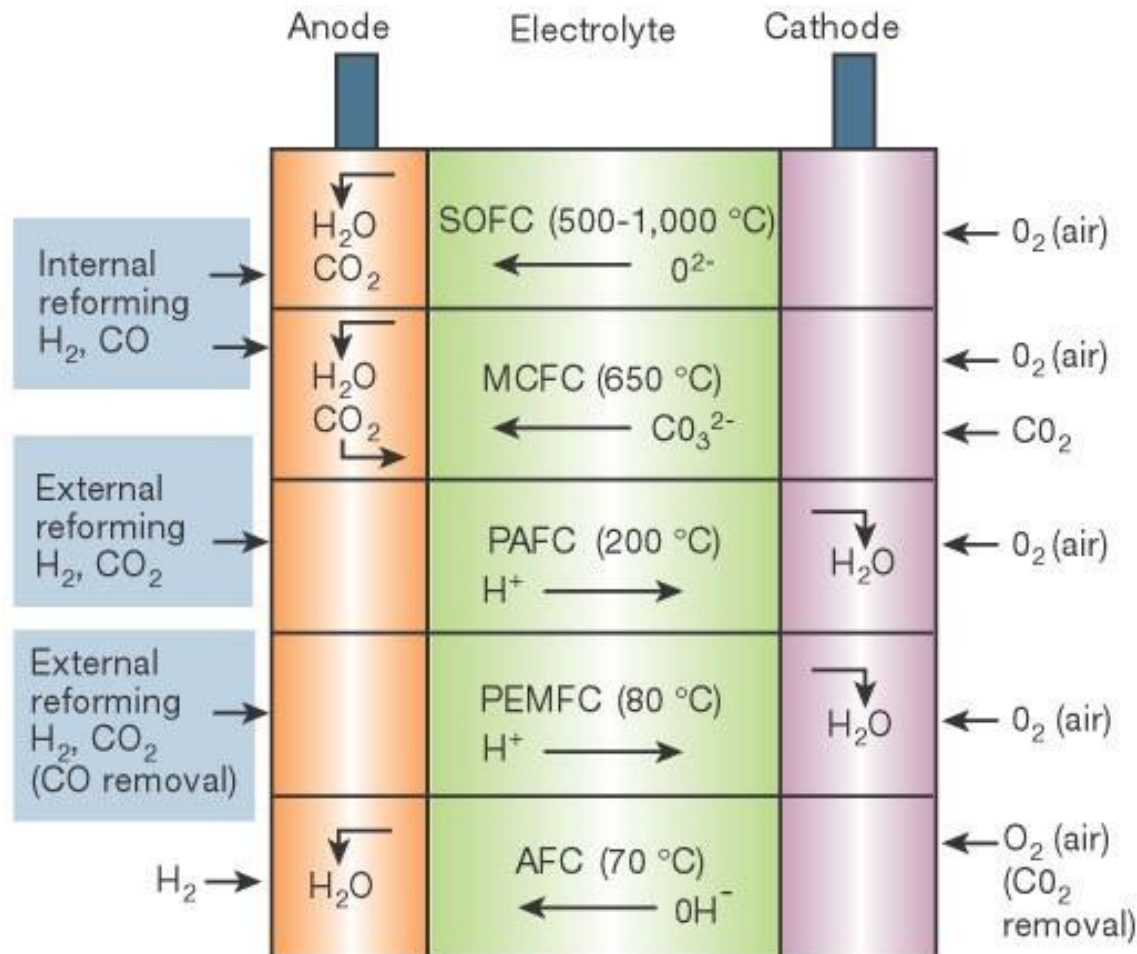
Rated power: ~ 25 W

Rated voltage: $\sim 28\text{--}36$ V

Rated power: ~ 1 kW

Fuel Cell

□ Types of Fuel Cell



Nature, **414** (2001) 345–352

Fuel Cell



□ Types of Fuel Cell

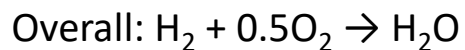
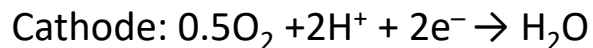
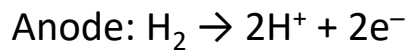
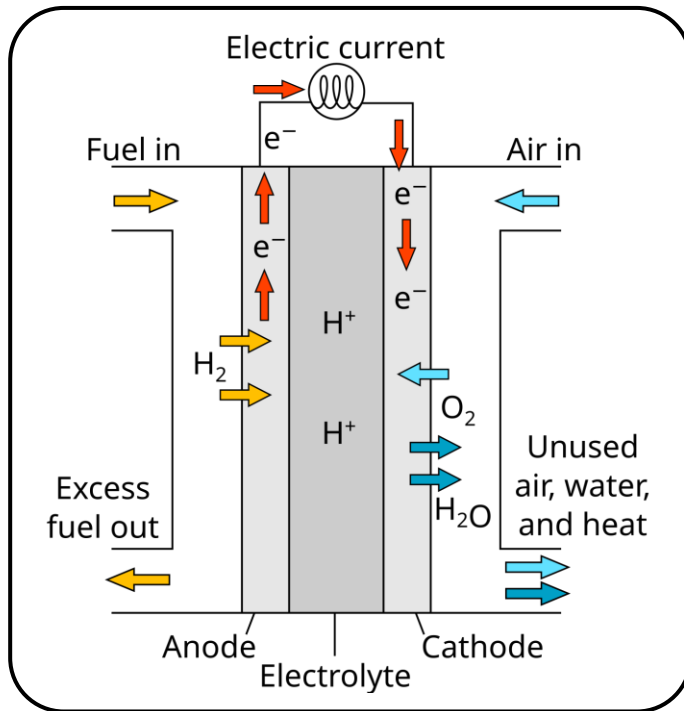
Type	Electrolyte	Stack Size	Eff (LHV)	Application	Advantages	Challenges
SOFC	Stabilized zirconia	1 kW–2 MW	45–65%	Auxiliary power Electric utility Distributed generation	High eff Fuel flexibility Solid electrolyte Suitable for CHP Hybrid/GT cycle	High temp corrosion Rapid degradation Long startup time
MCFC	Molten carbonate	300 kW–3 MW	45–60%	Electric utility Distributed generation	High eff Fuel flexibility Suitable for CHP Hybrid/GT cycle	High temp corrosion Rapid degradation Long startup time Low power density
PAFC	Phosphoric Acid	5–400 kW	35–42%	Distributed generation	Suitable for CHP Increased tolerance to fuel impurities	Expensive catalysts Long startup time Sulfur sensitivity
PEMFC	PFSA (Nafion)	<1 kW–0.1 MW	30–40%	Backup power Portable power Distributed generation Transportation Specialty vehicles	Low temp Low corrosion Electrolyte mgmt. Quick startup & shutdown	Expensive catalysts Sensitive to fuel impurities

Fuel Cell

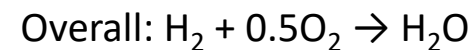
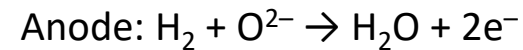
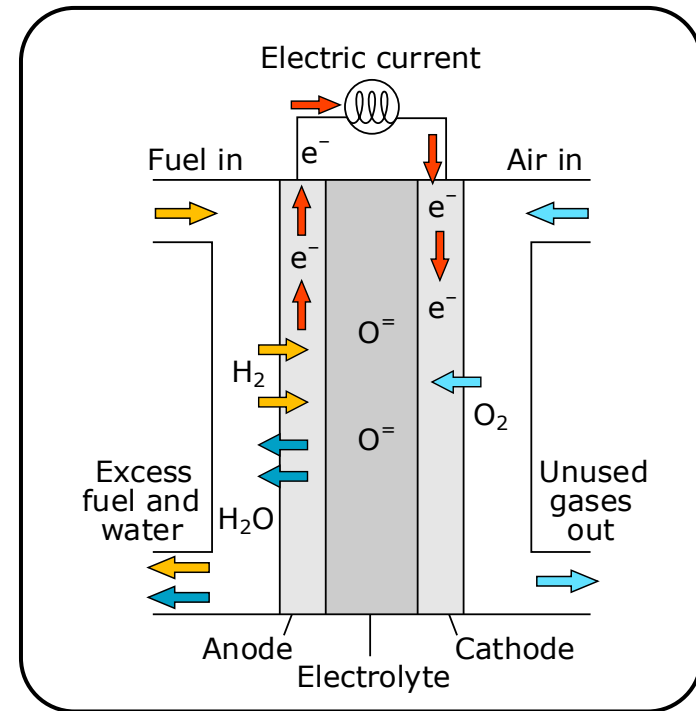


Types of Fuel Cell

● H⁺ Conducting Type (PEMFC)



● O²⁻ Conducting Type (SOFC)



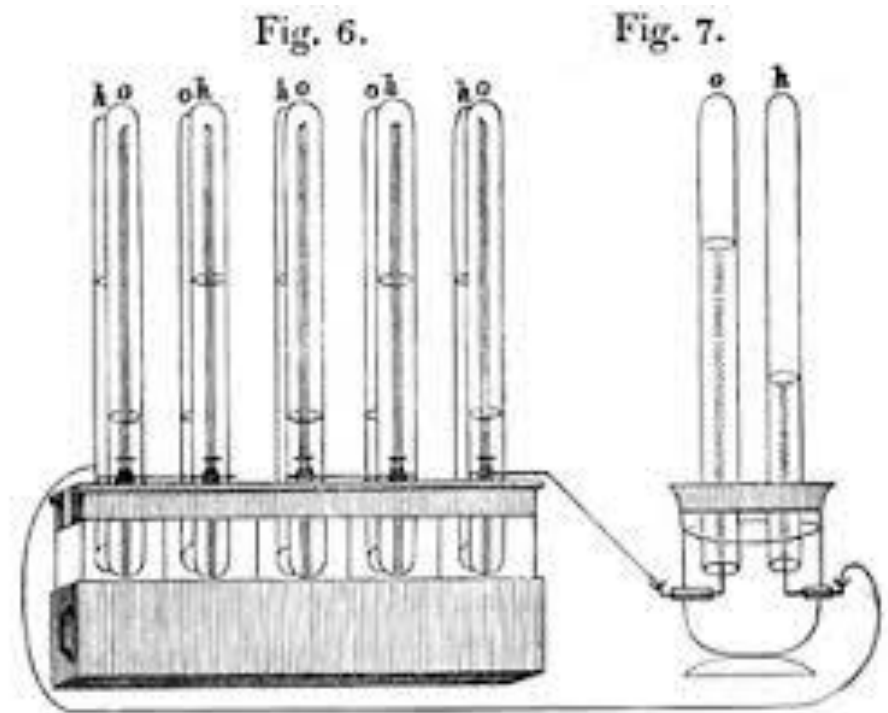
Fuel Cell

□ The Very First Fuel Cell

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**William Robert Grove**  
(1811–1896)



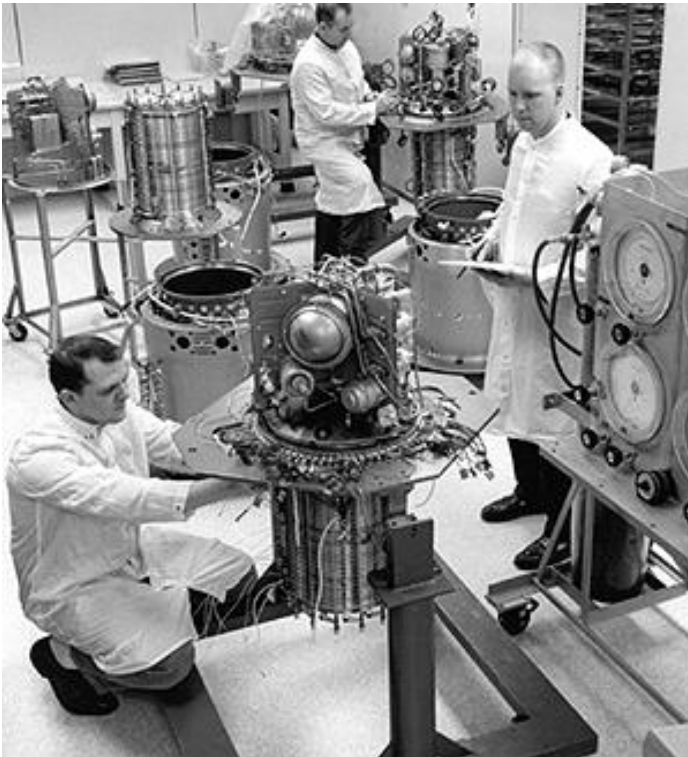
### The Grove Cell (1838)

Used a Pt electrode immersed in  $\text{HNO}_3$  and a Zn electrode in  $\text{ZnSO}_4$  to generate about 12 A of current at about 1.8 V

<Source: <http://americanhistory.si.edu/fuelcells/origins/orig1.htm>>

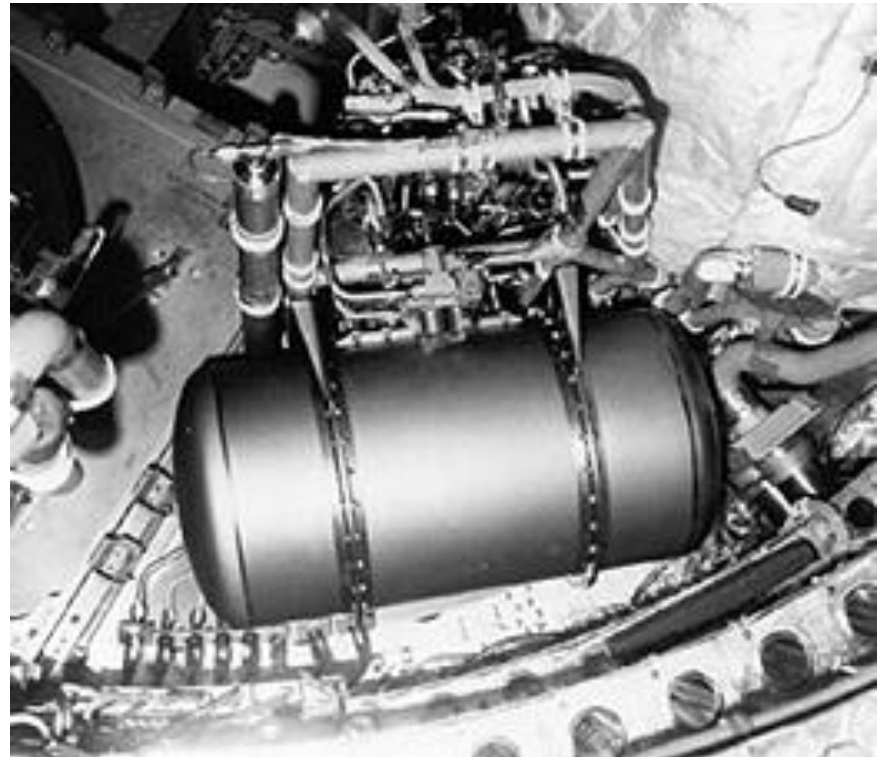
# Fuel Cell

## ❑ First Applied in 1960s Space Exploration



**Pratt & Whitney technicians assemble AFCs for Apollo service modules (1964)**

<Source: <https://americanhistory.si.edu/fuelcells/alk/alk3.htm> >

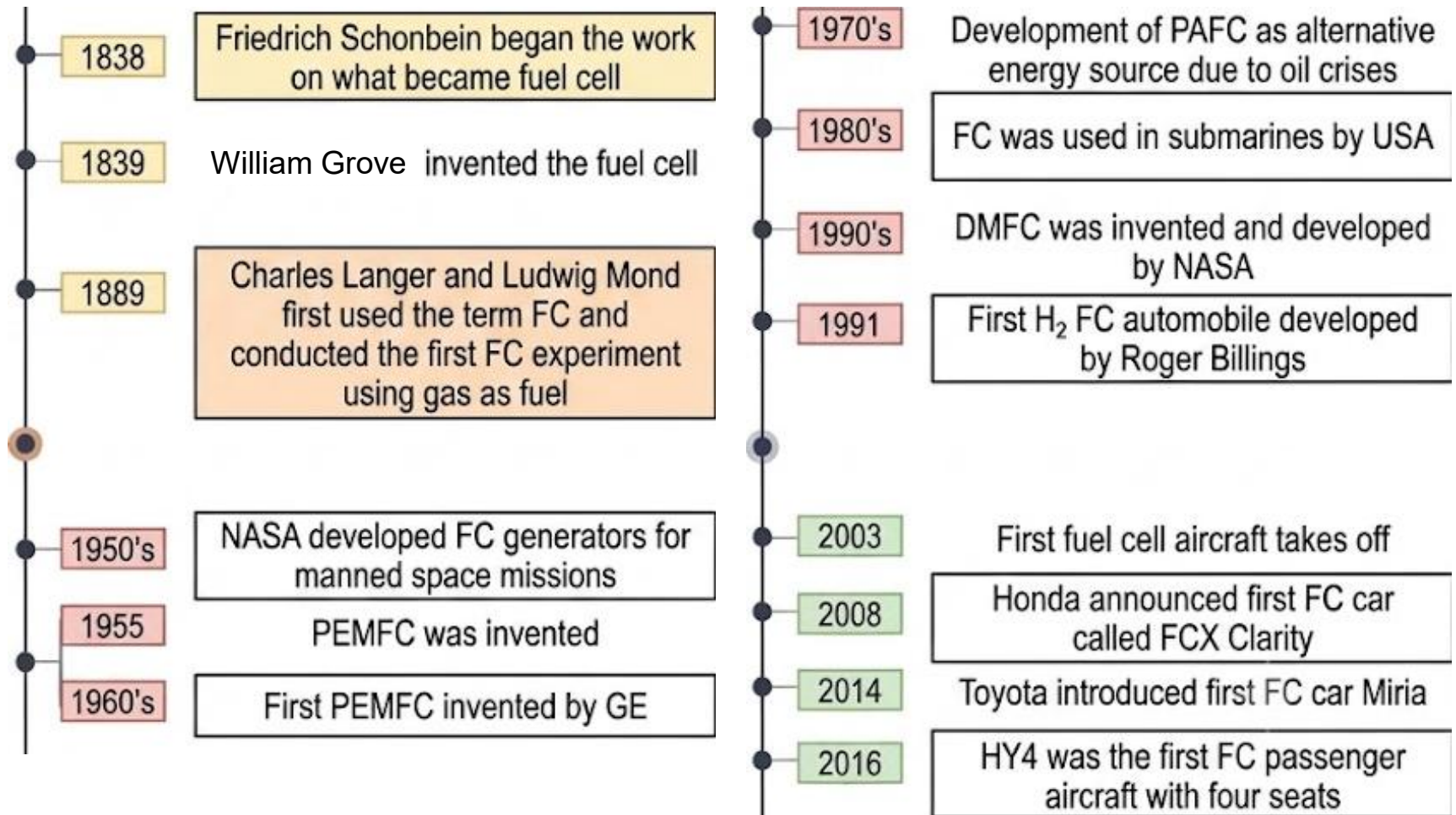


**PEMFC in the Gemini 7 spacecraft (1965)**

<Source: <https://americanhistory.si.edu/fuelcells/pem/pem3.htm> >

# Fuel Cell

## □ Timeline





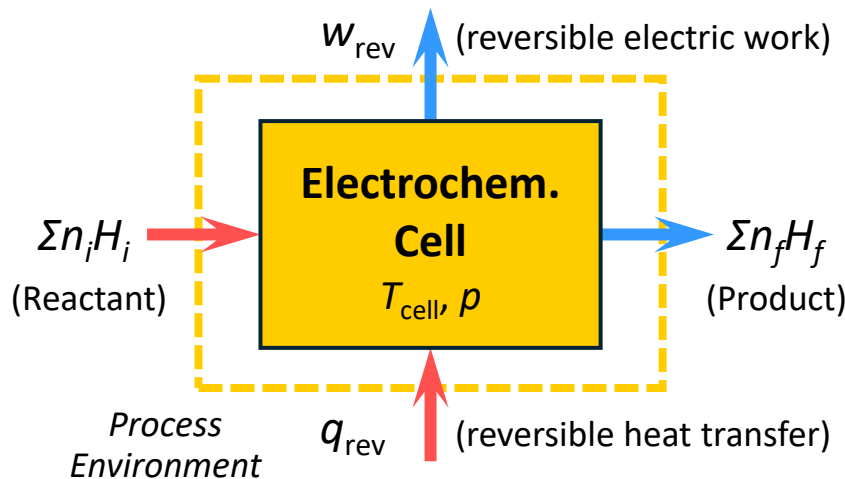
# Fuel Cell



## □ Gibbs Free Energy & Cell Voltage

The system's thermodynamic driving force to perform electrical work is quantified by the voltage

- In an equilibrium system with an isothermal reversible reaction ( $\text{H}_2 + 0.5\text{O}_2 \rightleftharpoons \text{H}_2\text{O}$ )



(1<sup>st</sup> law of TD)  $q_{\text{rev}} - w_{\text{rev}} = \Sigma n_f H_f - \Sigma n_i H_i = \Delta H$

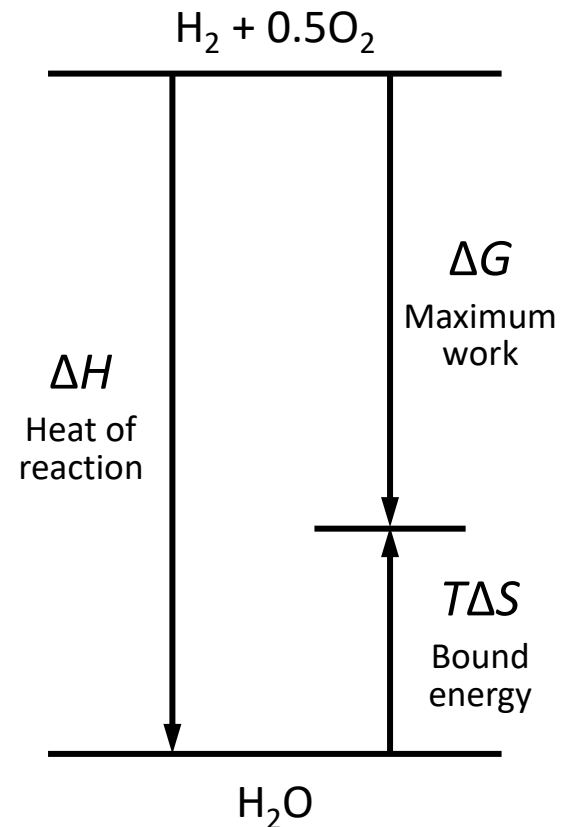
(2<sup>nd</sup> law of TD)  $\Delta S = q_{\text{rev}} / T_{\text{cell}}$   
 ( $\Delta H$ : enthalpy of the system  
 $\Delta S$ : entropy of the system)

Thus, the reversible work can be expressed by

$$w_{\text{rev}} = -\Delta H + T_{\text{cell}} \Delta S = -\Delta G$$

When  $w_{\text{rev}} > 0 \Leftrightarrow \Delta G < 0$ : Spontaneous (Fuel cell)

$w_{\text{rev}} < 0 \Leftrightarrow \Delta G > 0$ : Nonspontaneous (Electrolyzer)





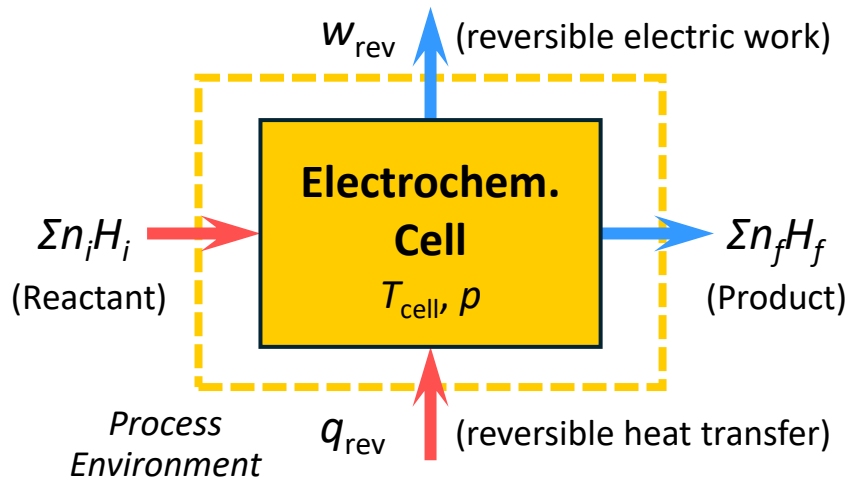
# Fuel Cell



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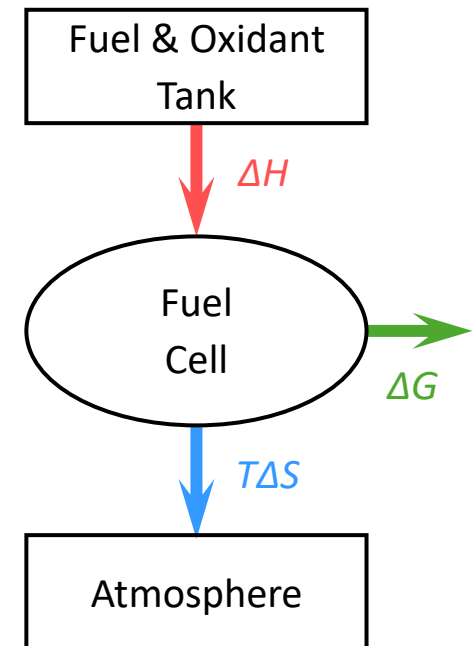
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$$\epsilon_{\text{FC,th}} = 1 - \frac{T\Delta S}{\Delta H}$$

~80% @ 600 °C  
 ~70% @ 1000 °C

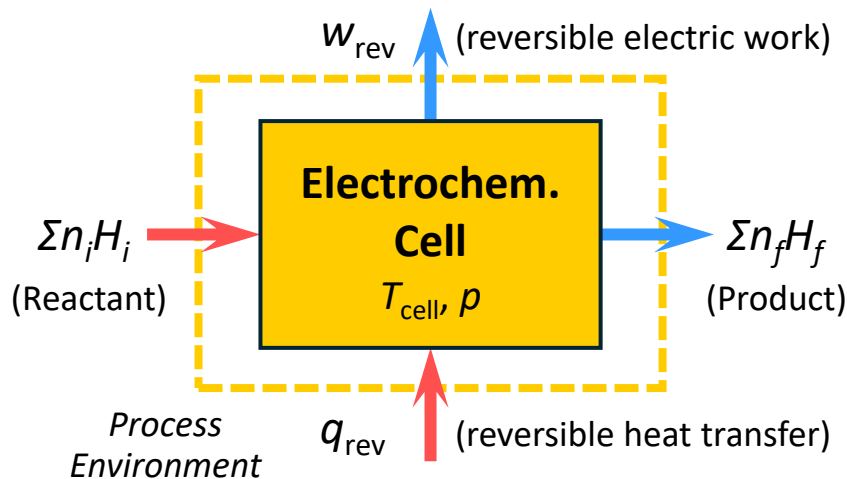
# Fuel Cell



## □ Gibbs Free Energy & Cell Voltage

The system's thermodynamic driving force to perform electrical work is quantified by the voltage

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(1<sup>st</sup> law of TD)  $q_{\text{rev}} - w_{\text{rev}} = \Sigma n_f H_f - \Sigma n_i H_i = \Delta H$

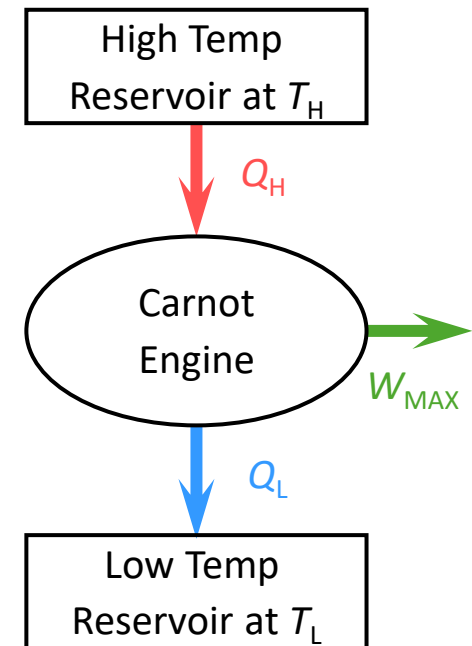
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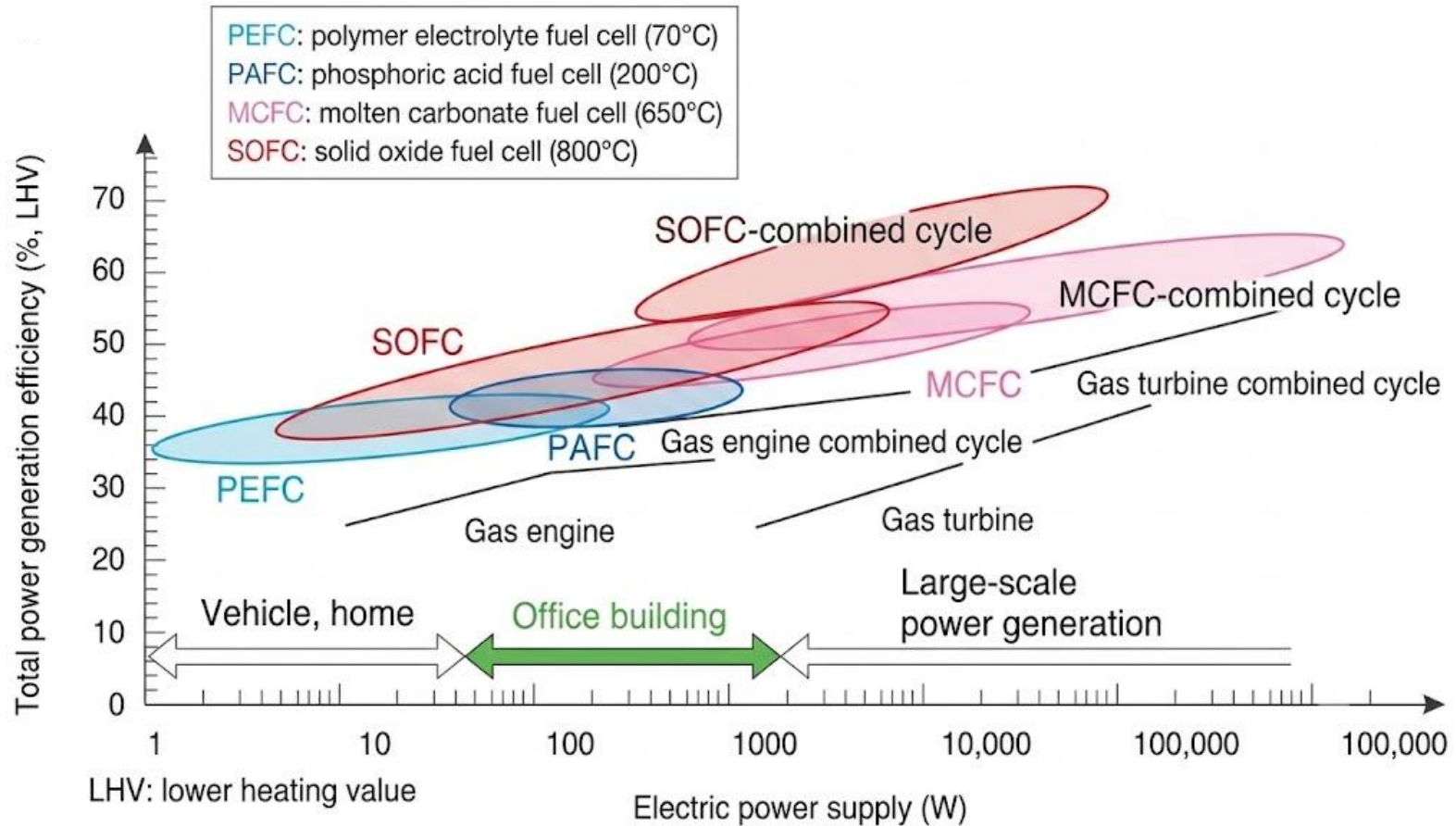


$$\varepsilon_{\text{Carnot}} = 1 - \frac{Q_L}{Q_H}$$

~65% @ 600 °C  
 ~77% @ 1000 °C

# Fuel Cell

## □ Efficiency Comparison



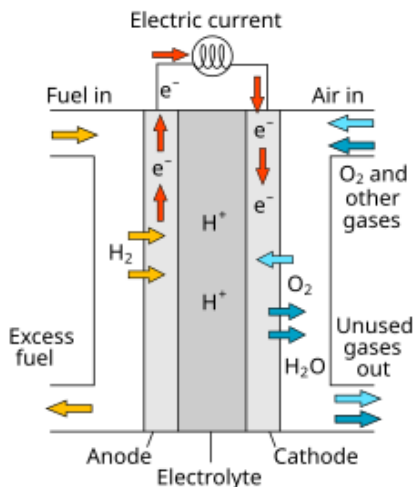
# Fuel Cell



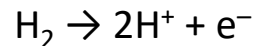
## □ Gibbs Free Energy & Cell Voltage

The system's thermodynamic driving force to perform electrical work is quantified by the voltage

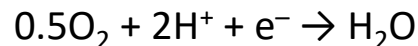
- The electrical effect can be explained by thermodynamic principles



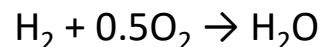
(Anode)



(Cathode)



(Overall)



The electric current  $I$  is expressed as,

$$I = \dot{q} = \dot{N}_e e = (\dot{n}_e - N_A) e = 2\dot{n}_{\text{H}_2} F$$

The electric power  $P_{\text{rev}}$  is the same as the thermodynamic quantity, so

$$P_{\text{rev}} = V_{\text{rev}} I = \dot{n}_{\text{H}_2} w_{\text{rev}} = \dot{n}_{\text{H}_2} (-\Delta G)$$

Hence, the reversible voltage  $V_{\text{rev}}$  is

$$\therefore V_{\text{rev}} = -\frac{\Delta G}{2F}$$

Under the assumption of ideal gas

$$\Delta G(T, p) = \Delta H(T) - T\Delta S(T, p),$$

where

$$S(T, p) = S^\circ + \int_{T_0}^T \frac{C_p(t)}{t} dt - R \ln \left( \frac{p}{p_0} \right)$$

Then

$$\Delta G(T, p) = \Delta G^\circ + RT \ln \left( \frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2} p_{\text{O}_2}^{0.5}} \right)$$

Thus, the electromotive force  $V_{\text{rev}}$  is

$$\therefore V_{\text{rev}}(T, p) = -\frac{\Delta G^\circ(T)}{2F} - \frac{RT}{2F} \ln \left( \frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2} p_{\text{O}_2}^{0.5}} \right)$$

(e.g.,  $\Delta G^\circ = -237.14 \text{ kJ mol}^{-1} \Rightarrow V_{\text{rev}}^\circ \approx 1.23 \text{ V}$  at  $25^\circ\text{C}$ ,  $1 \text{ atm}$ )

# Quiz



## EV vs. FCV

Discuss the differences between electric and fuel-cell vehicles

**End of Document**